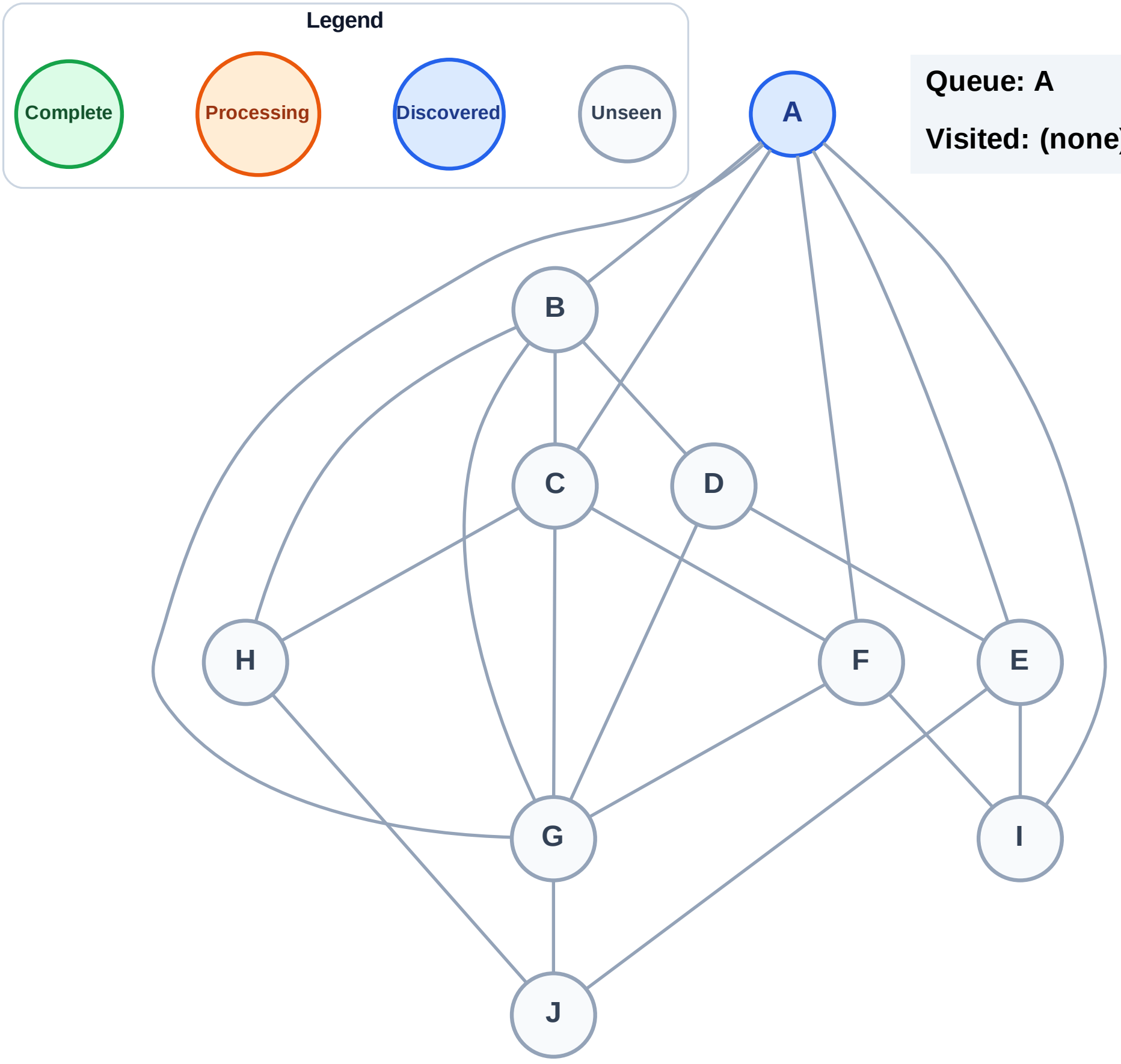
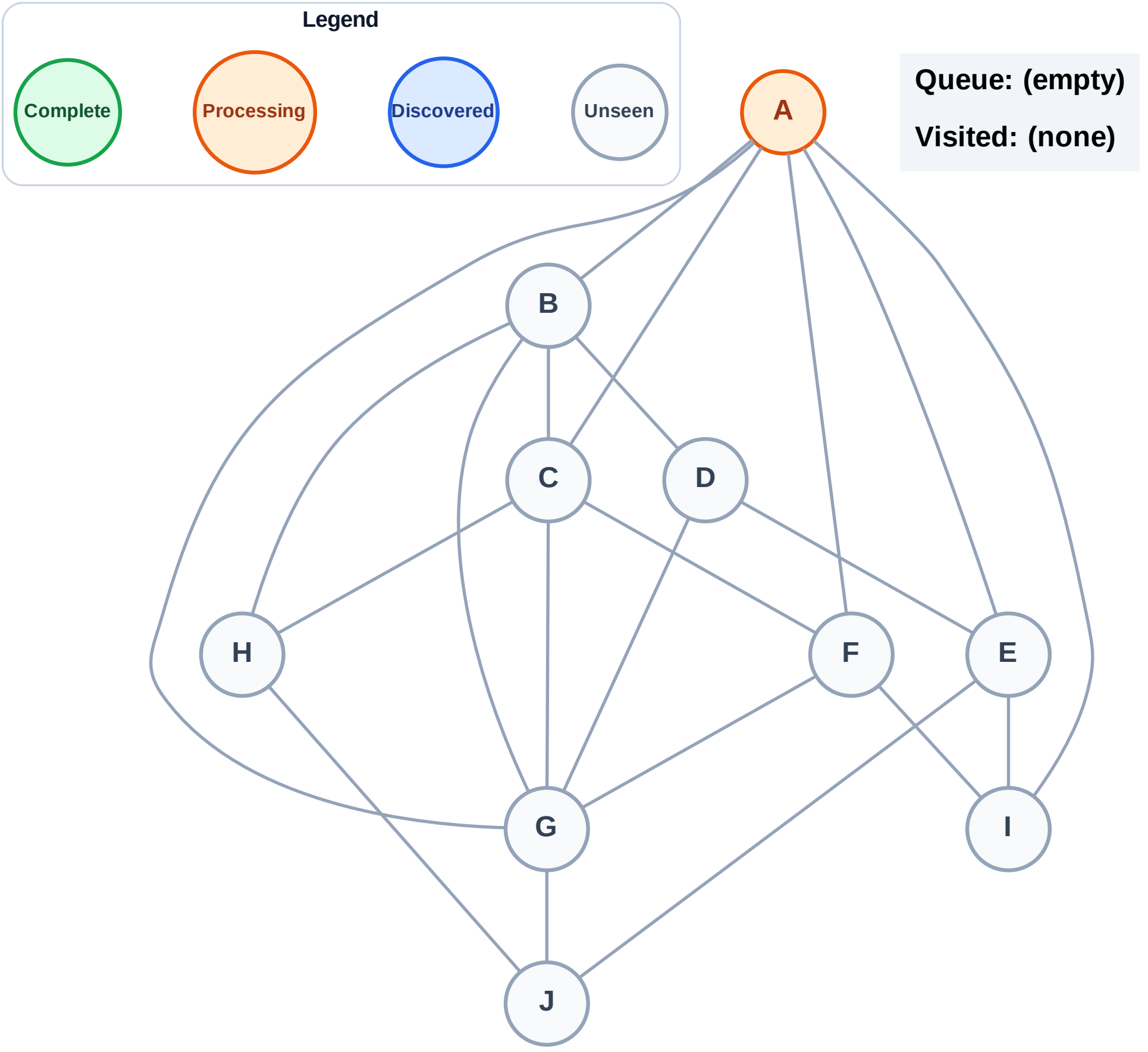
BFS Traversal

Step 1: Start at A



BFS Traversal

Step 2: Process node A

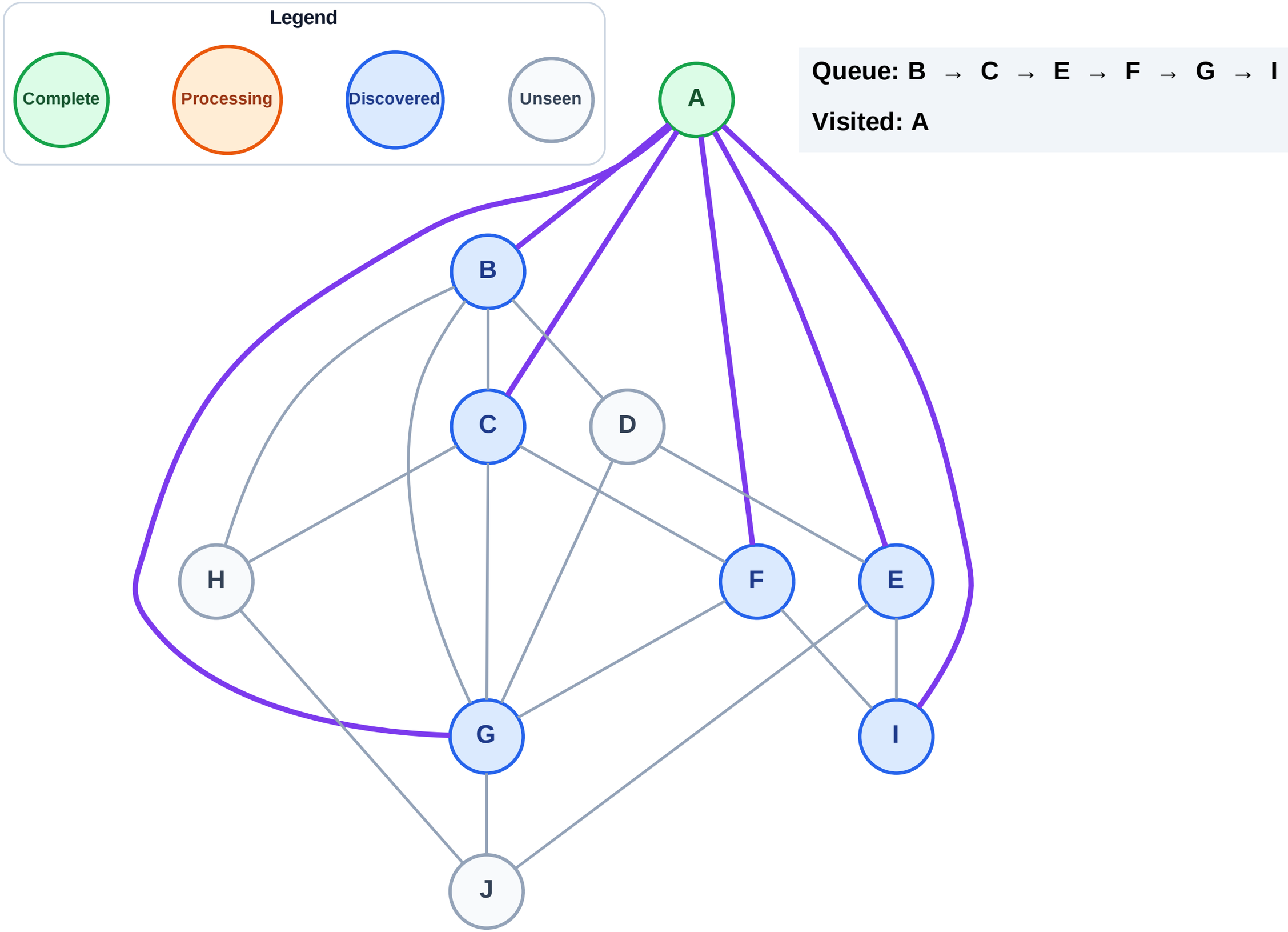


Queue: (empty)

Visited: (none)

BFS Traversal

Step 3: Discover B, C, E, F, G, I from A

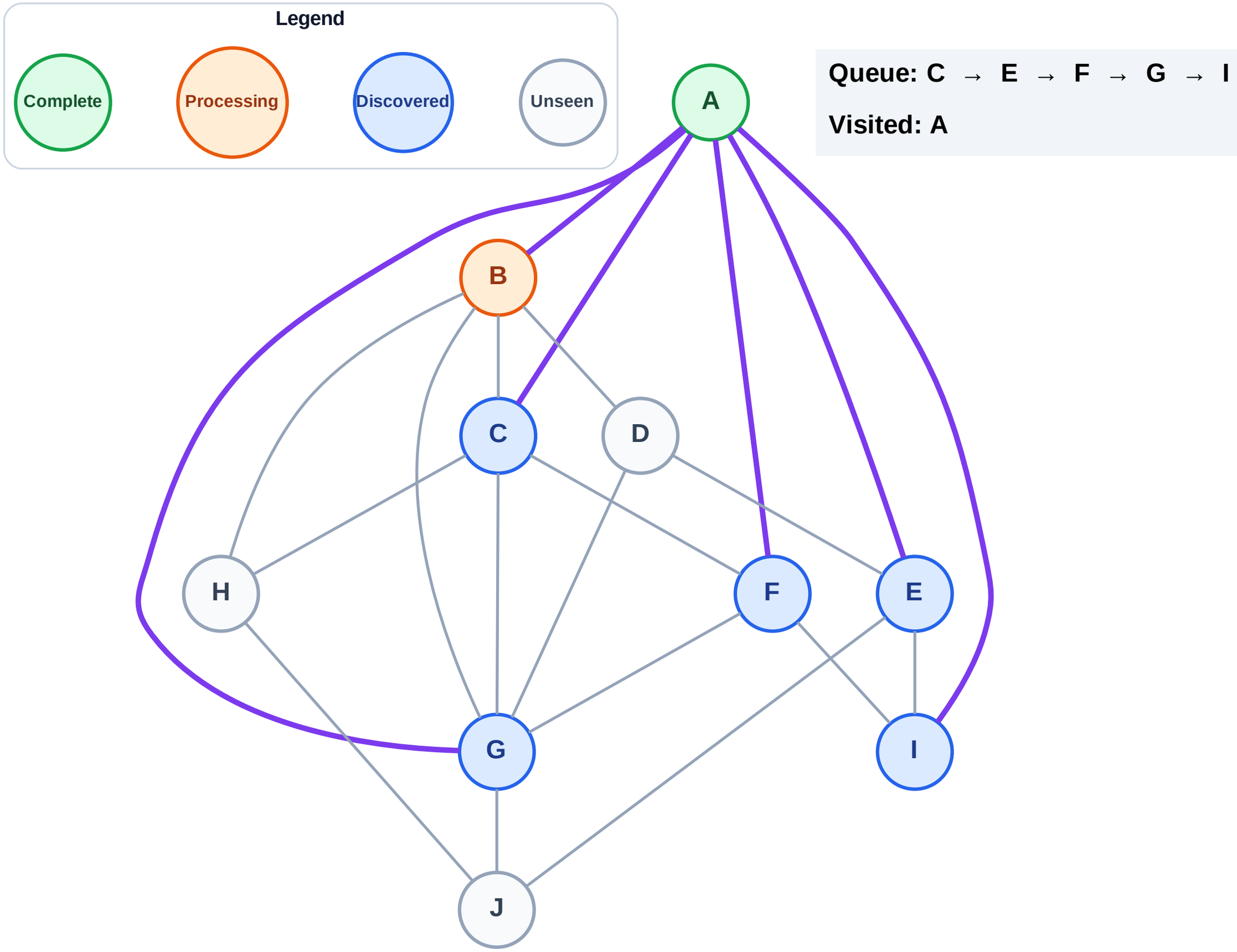


Queue: B → C → E → F → G → I

Visited: A

BFS Traversal

Step 4: Process node B

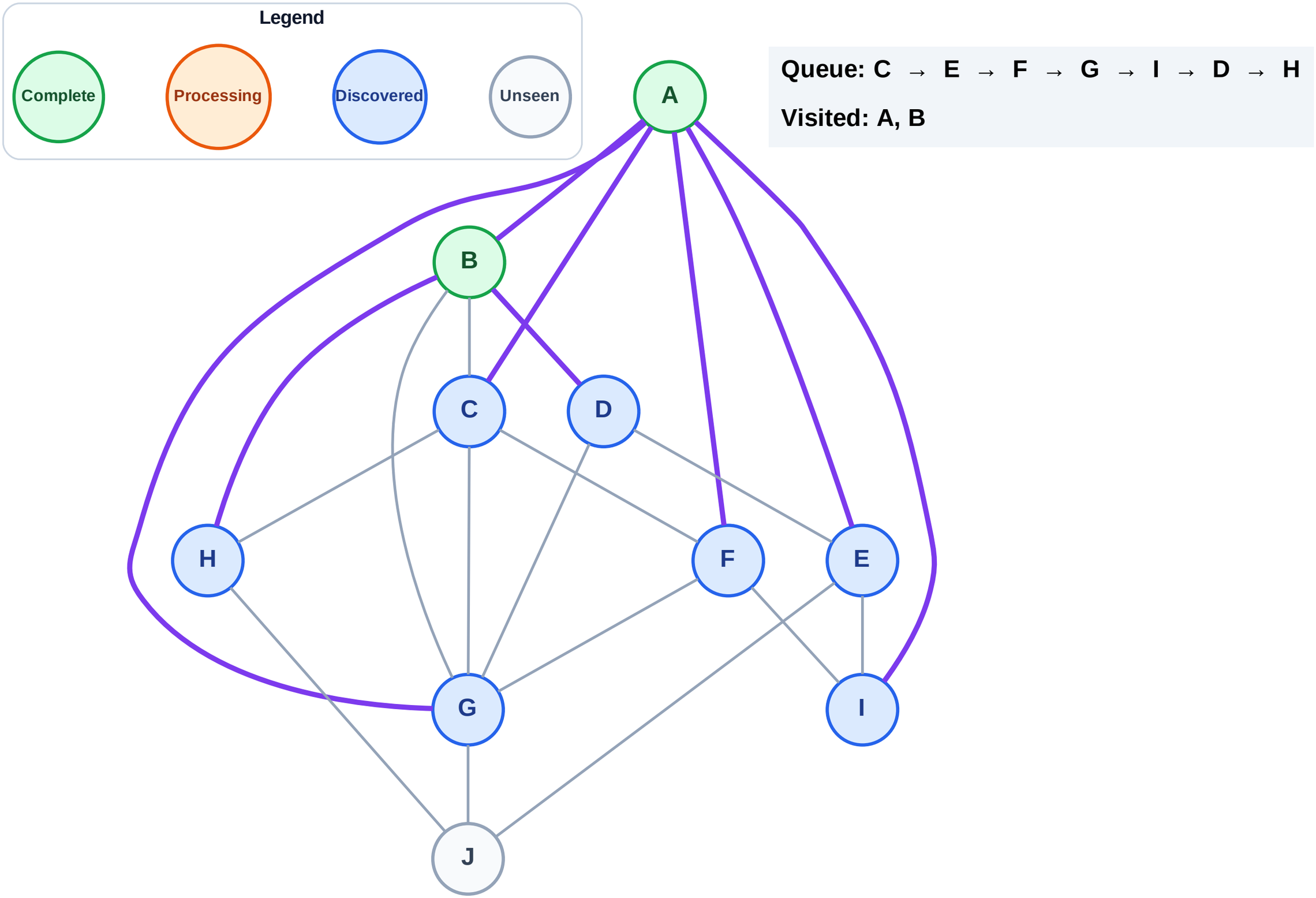


Queue: C → E → F → G → I

Visited: A

BFS Traversal

Step 5: Discover D, H from B

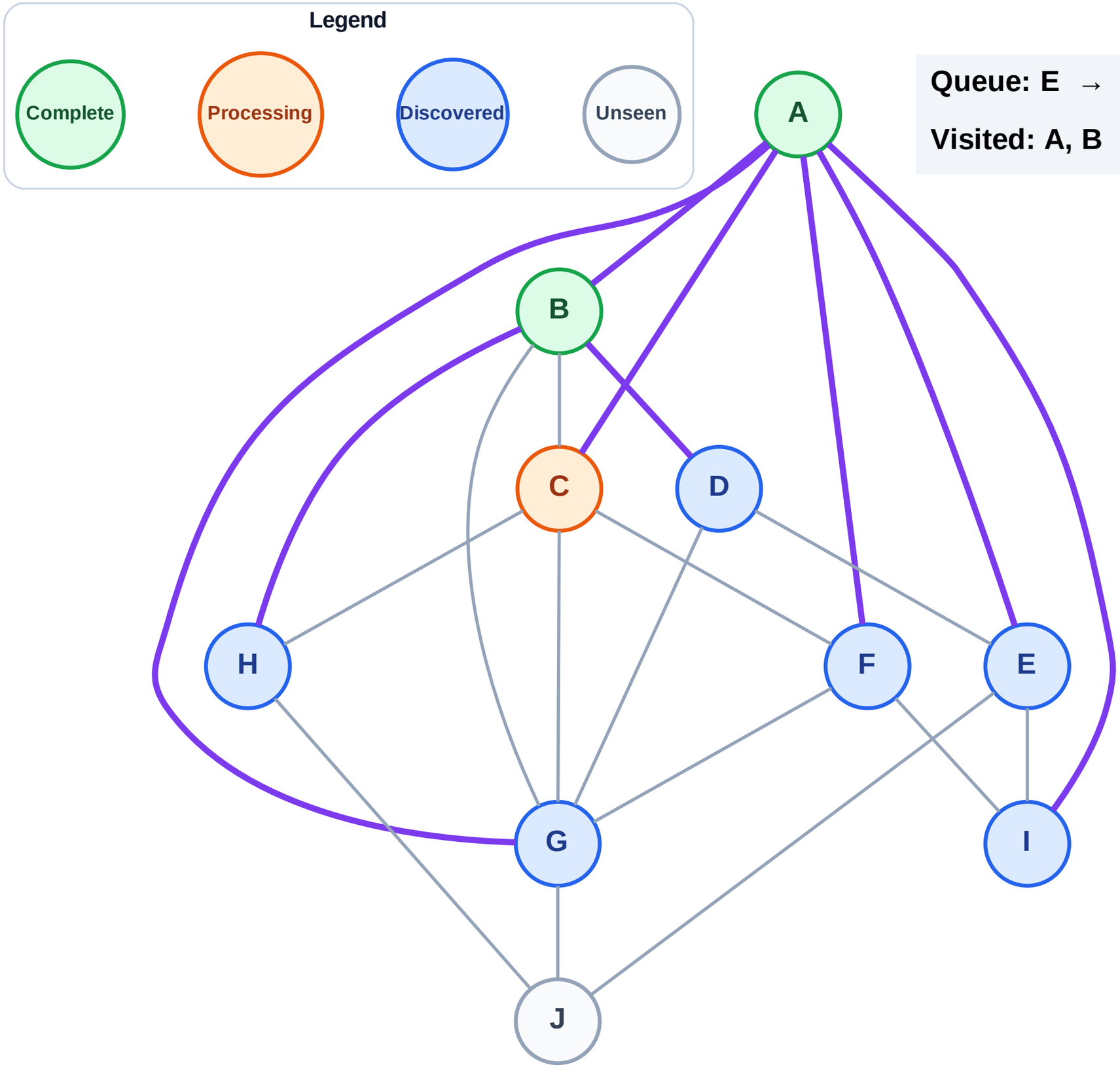


Queue: C → E → F → G → I → D → H

Visited: A, B

BFS Traversal

Step 6: Process node C

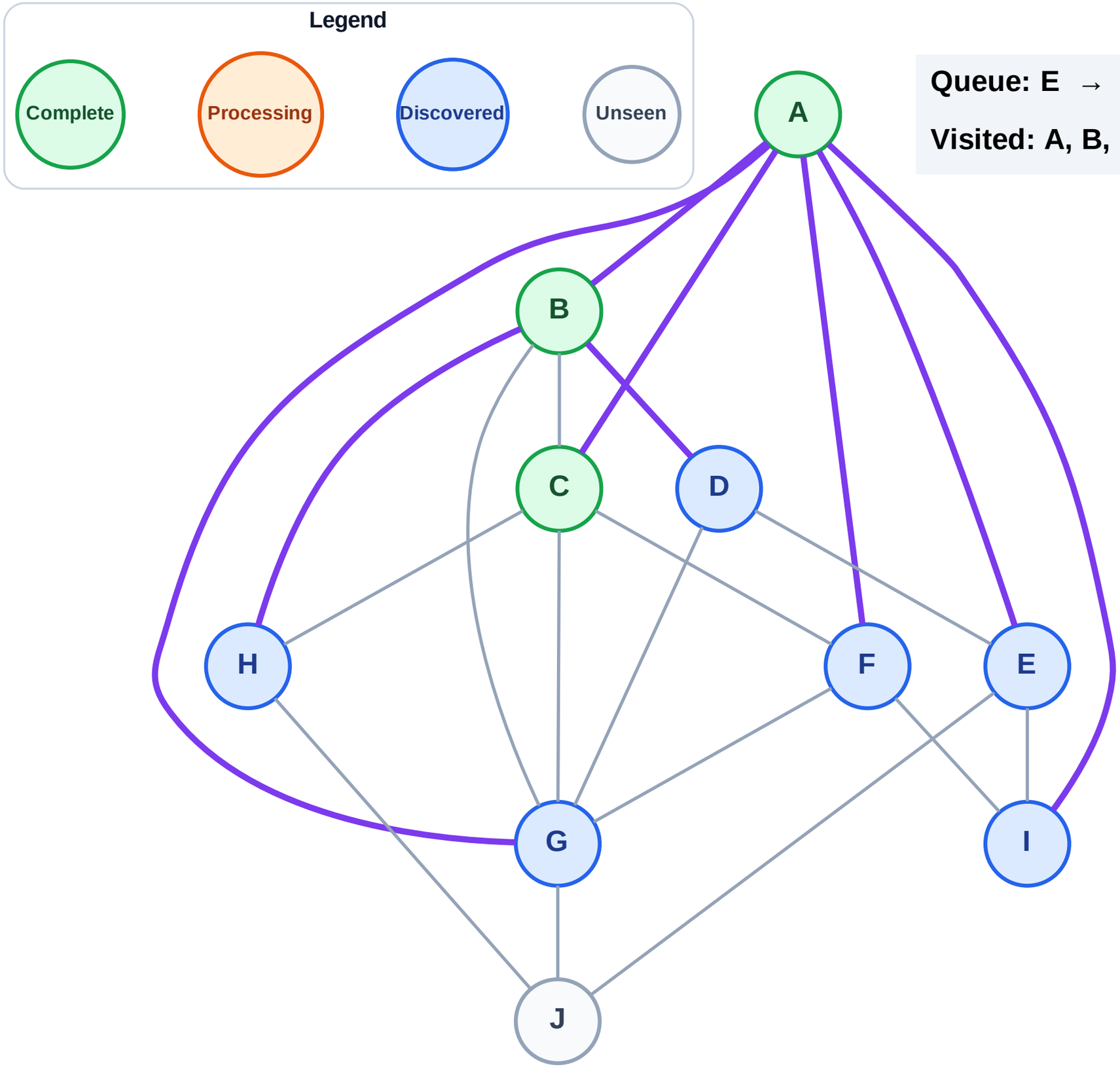


Queue: E → F → G → I → D → H

Visited: A, B

BFS Traversal

Step 7: No new neighbors from C

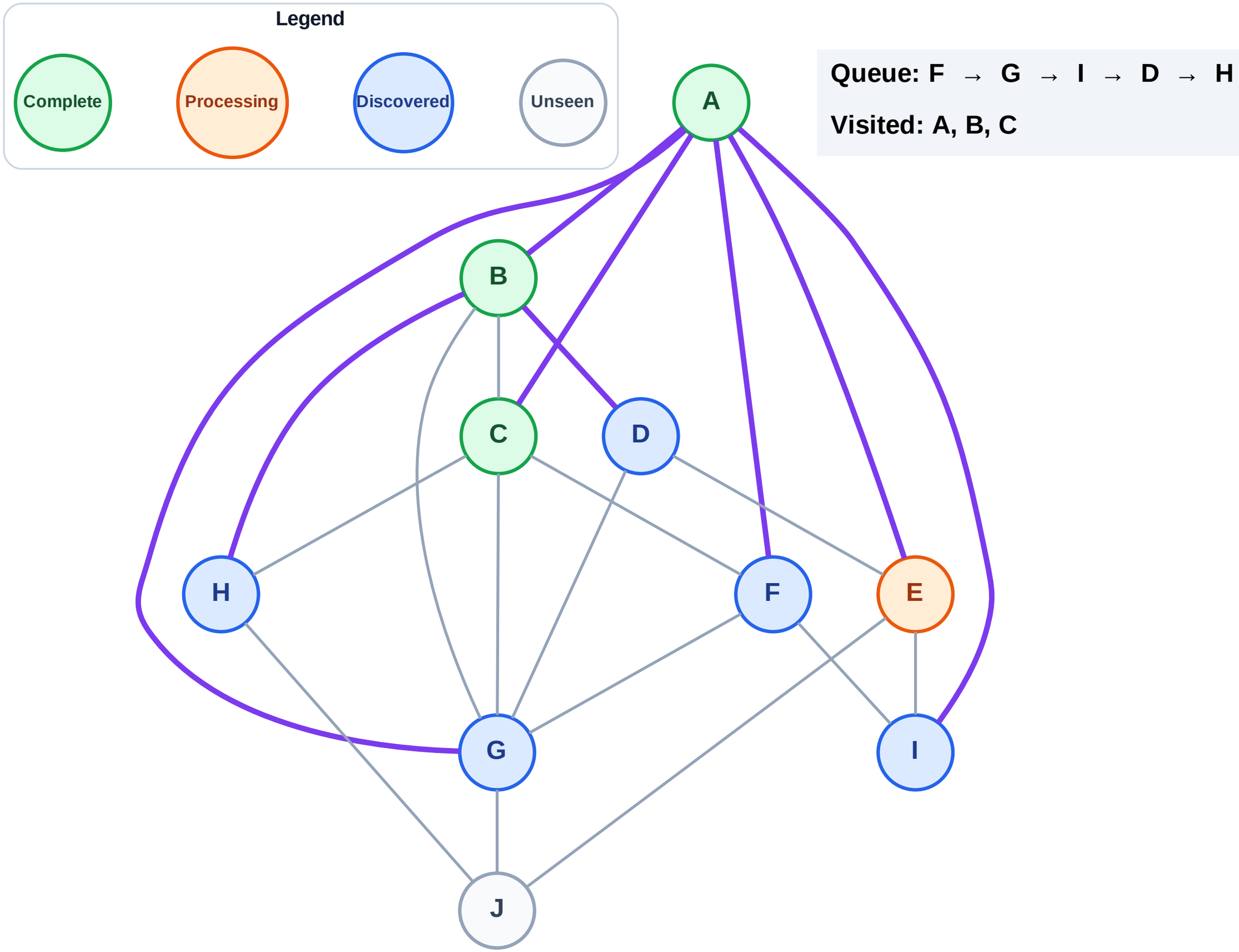


Queue: E → F → G → I → D → H

Visited: A, B, C

BFS Traversal

Step 8: Process node E

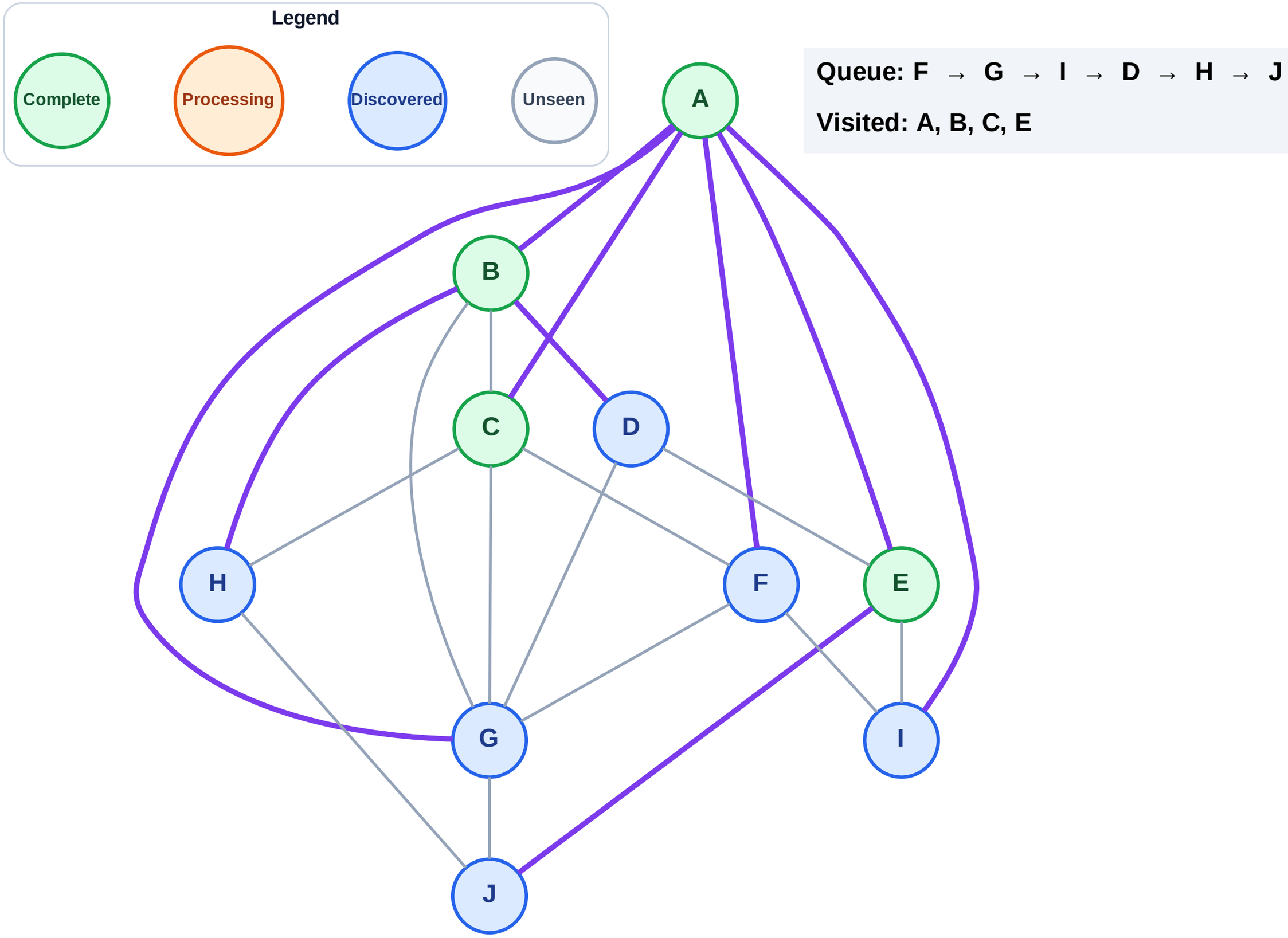


Queue: F → G → I → D → H

Visited: A, B, C

BFS Traversal

Step 9: Discover J from E

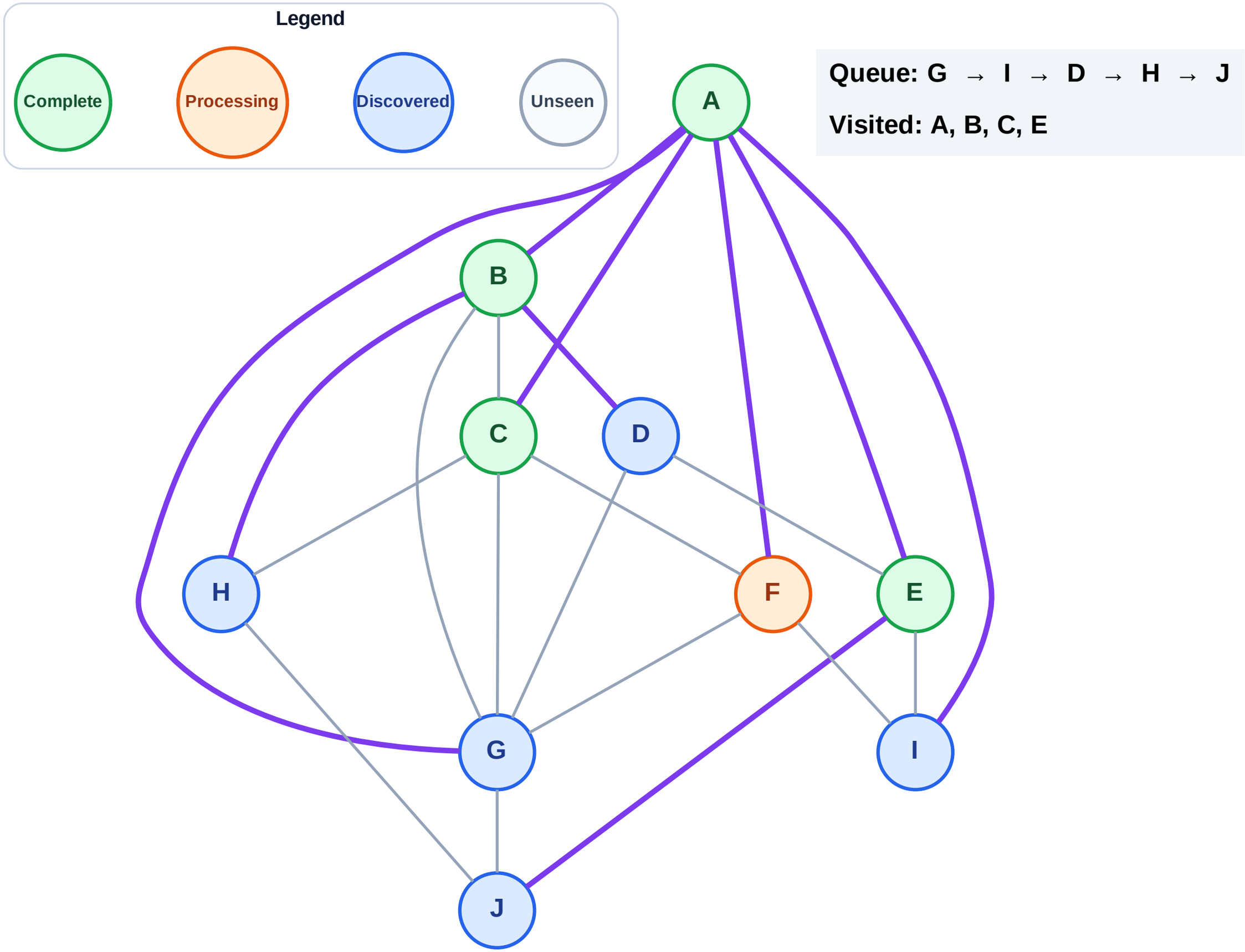


Queue: F → G → I → D → H → J

Visited: A, B, C, E

BFS Traversal

Step 10: Process node F

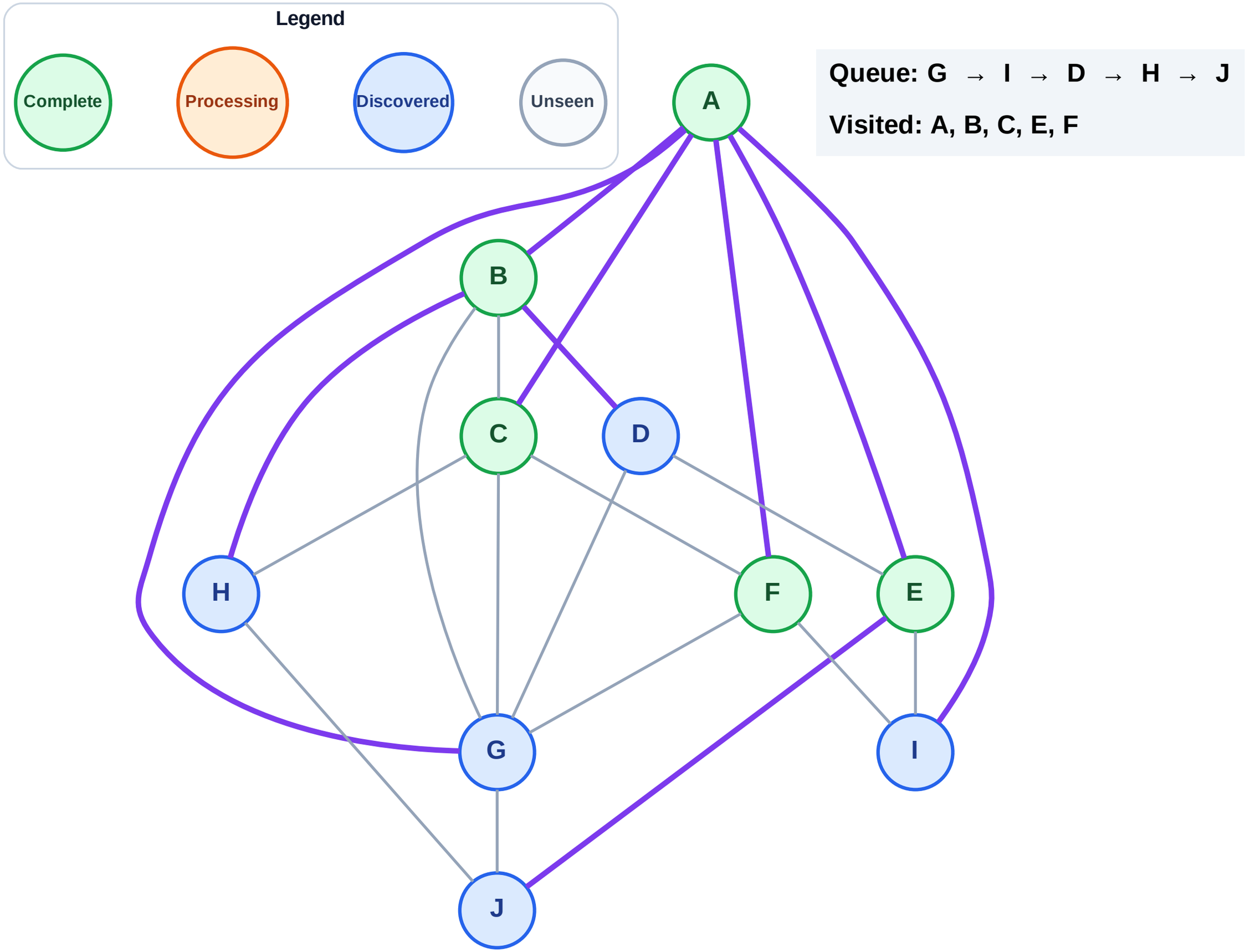


Queue: G → I → D → H → J

Visited: A, B, C, E

BFS Traversal

Step 11: No new neighbors from F

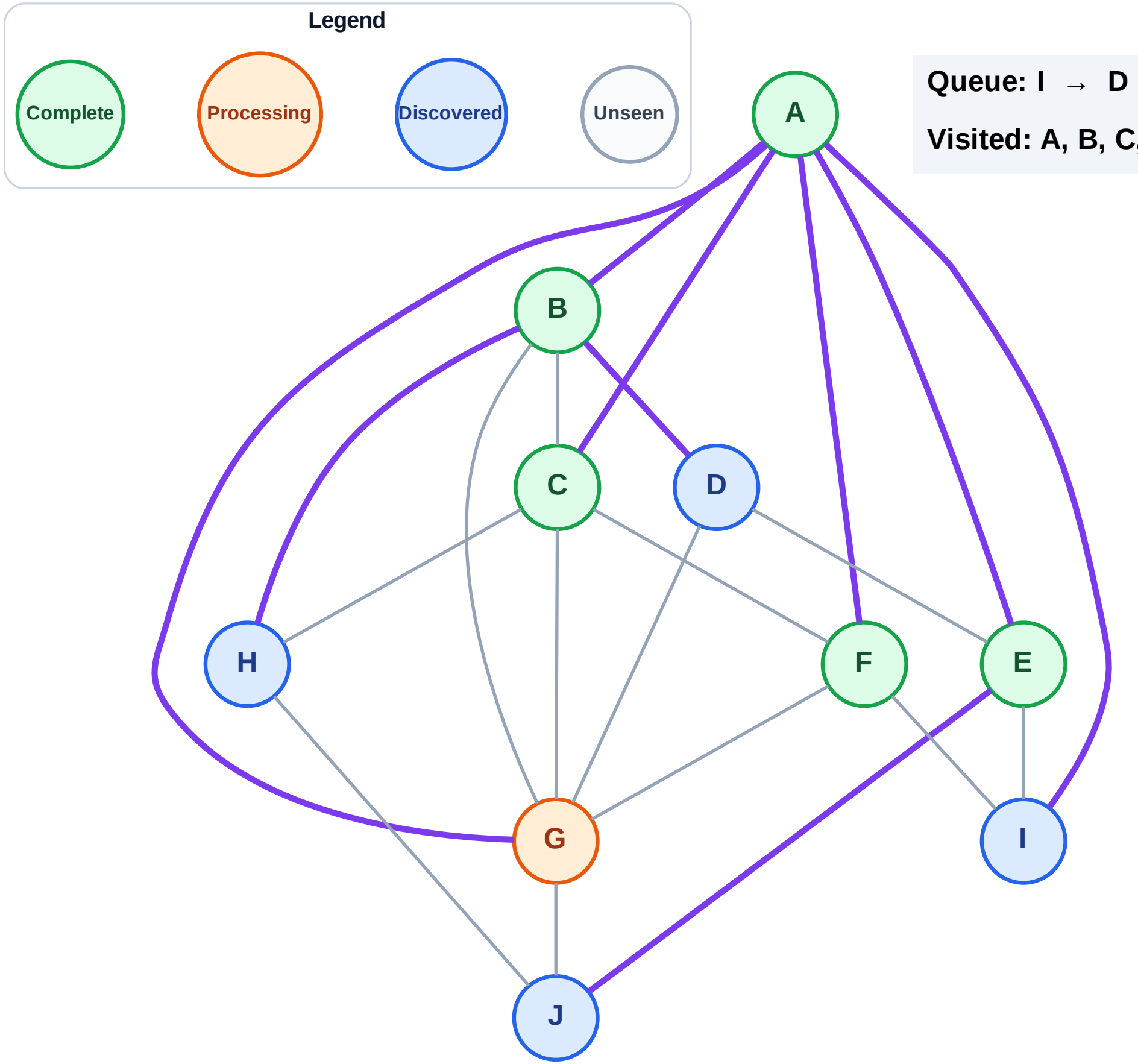


Queue: G → I → D → H → J

Visited: A, B, C, E, F

BFS Traversal

Step 12: Process node G

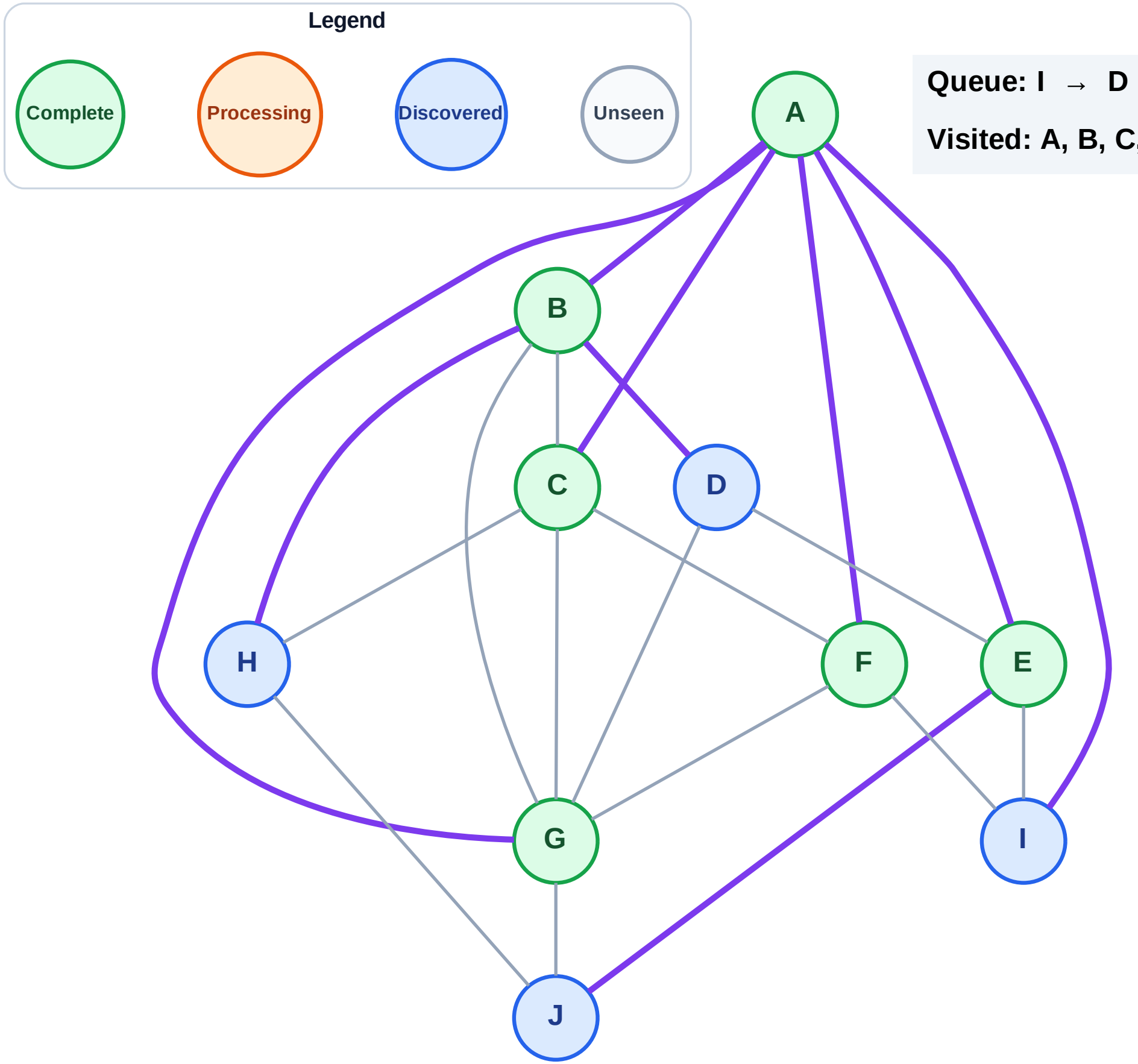


Queue: I → D → H → J

Visited: A, B, C, E, F

BFS Traversal

Step 13: No new neighbors from G

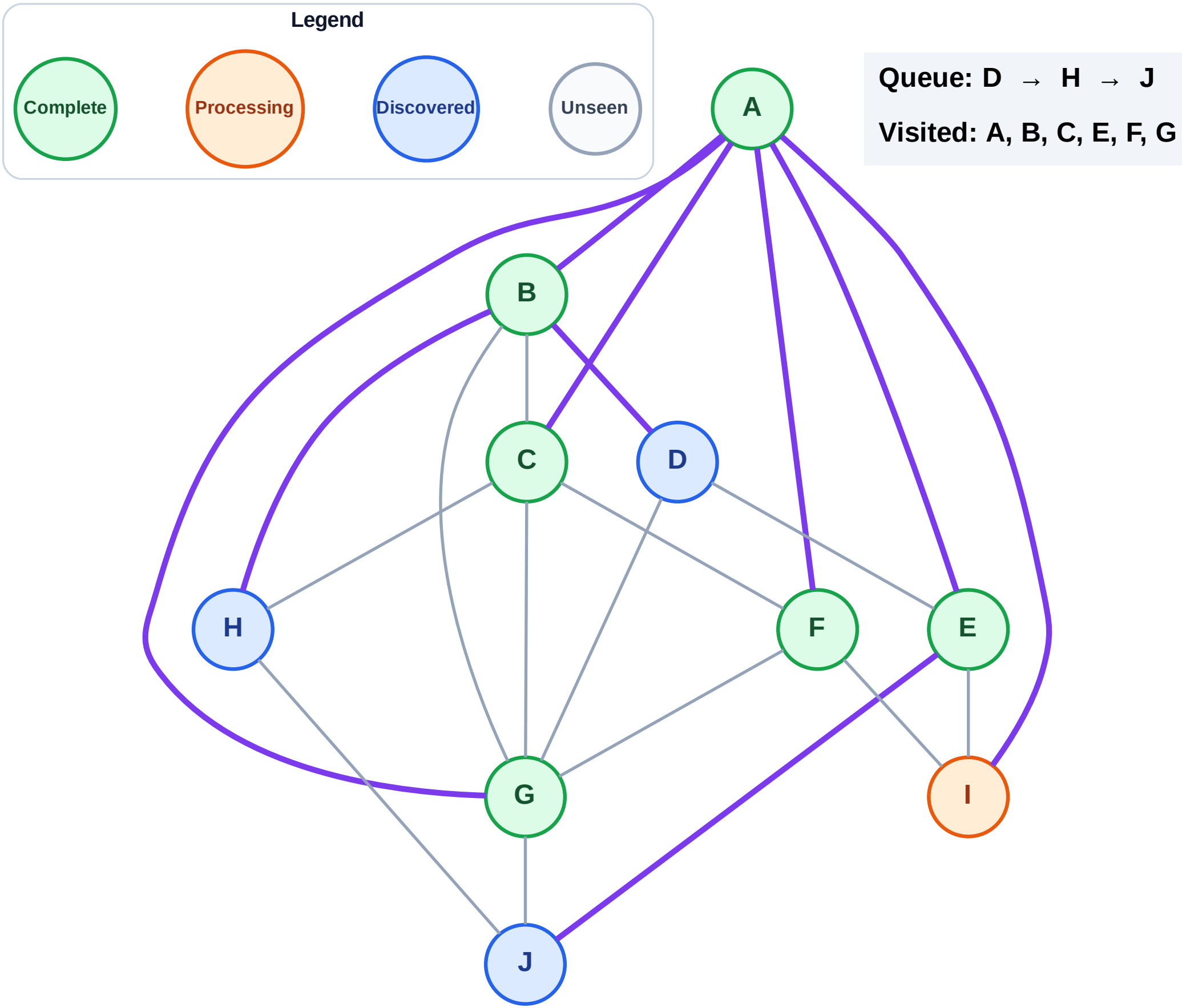


Queue: I → D → H → J

Visited: A, B, C, E, F, G

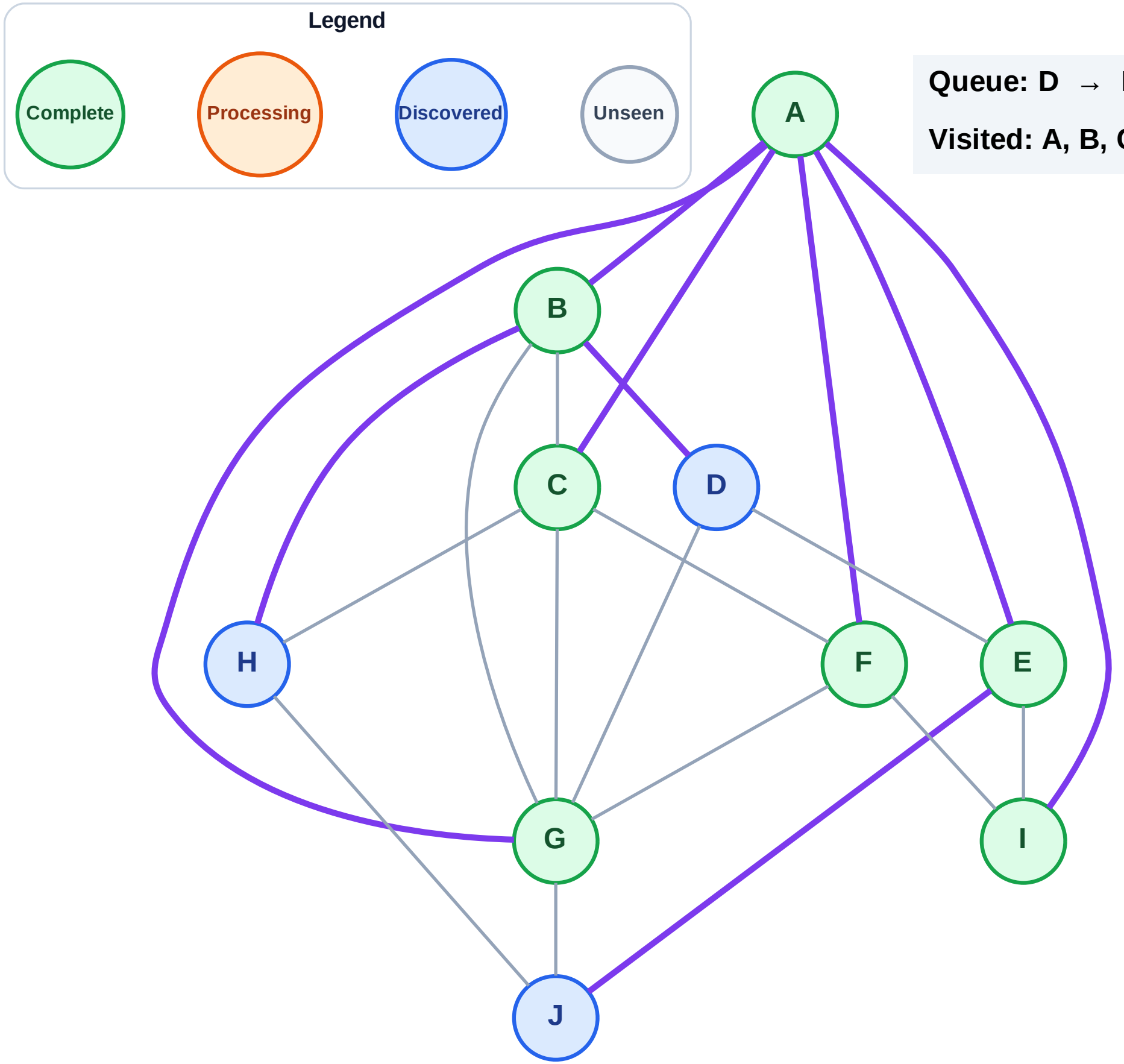
BFS Traversal

Step 14: Process node I



BFS Traversal

Step 15: No new neighbors from I

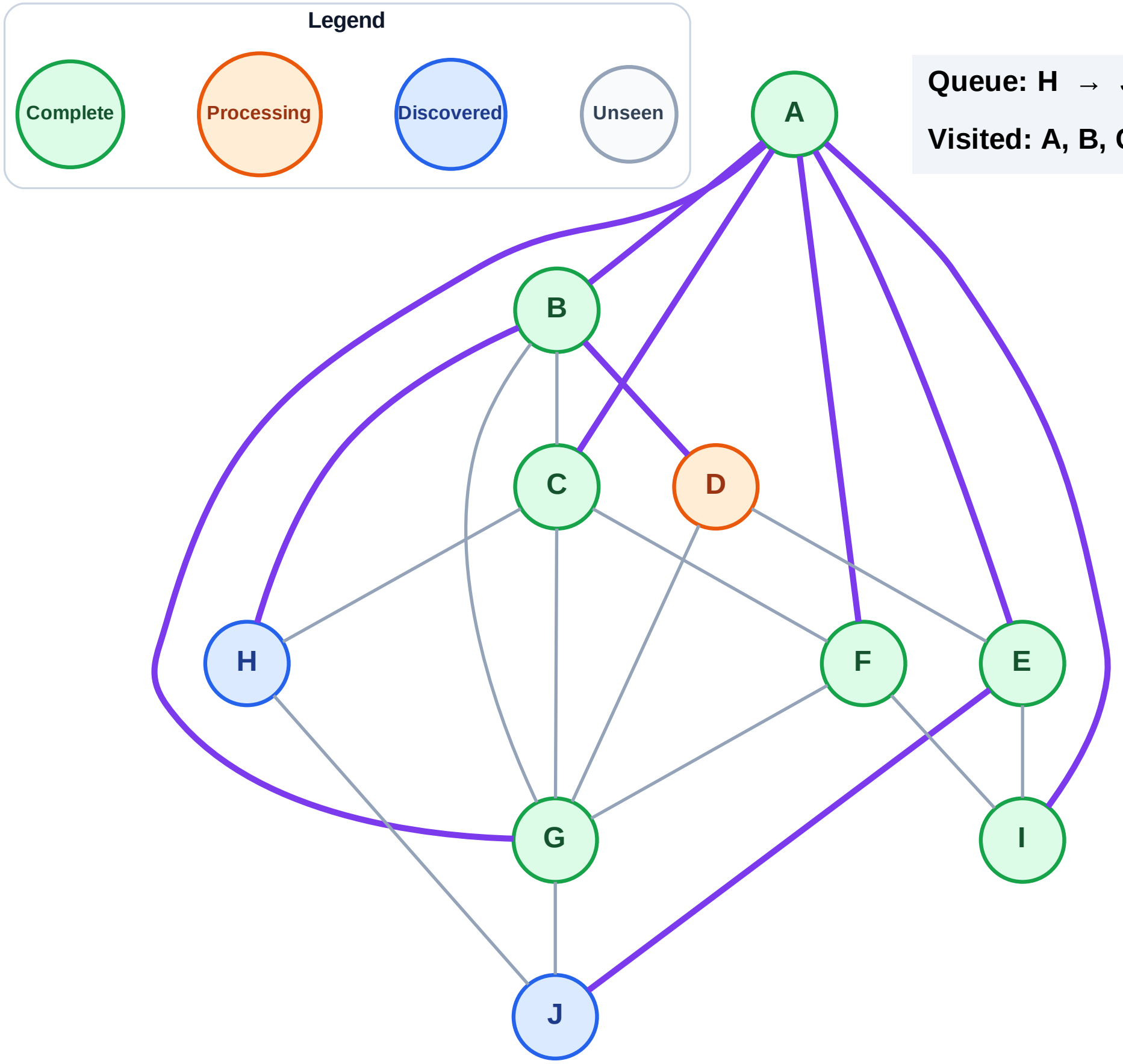


Queue: D → H → J

Visited: A, B, C, E, F, G, I

BFS Traversal

Step 16: Process node D

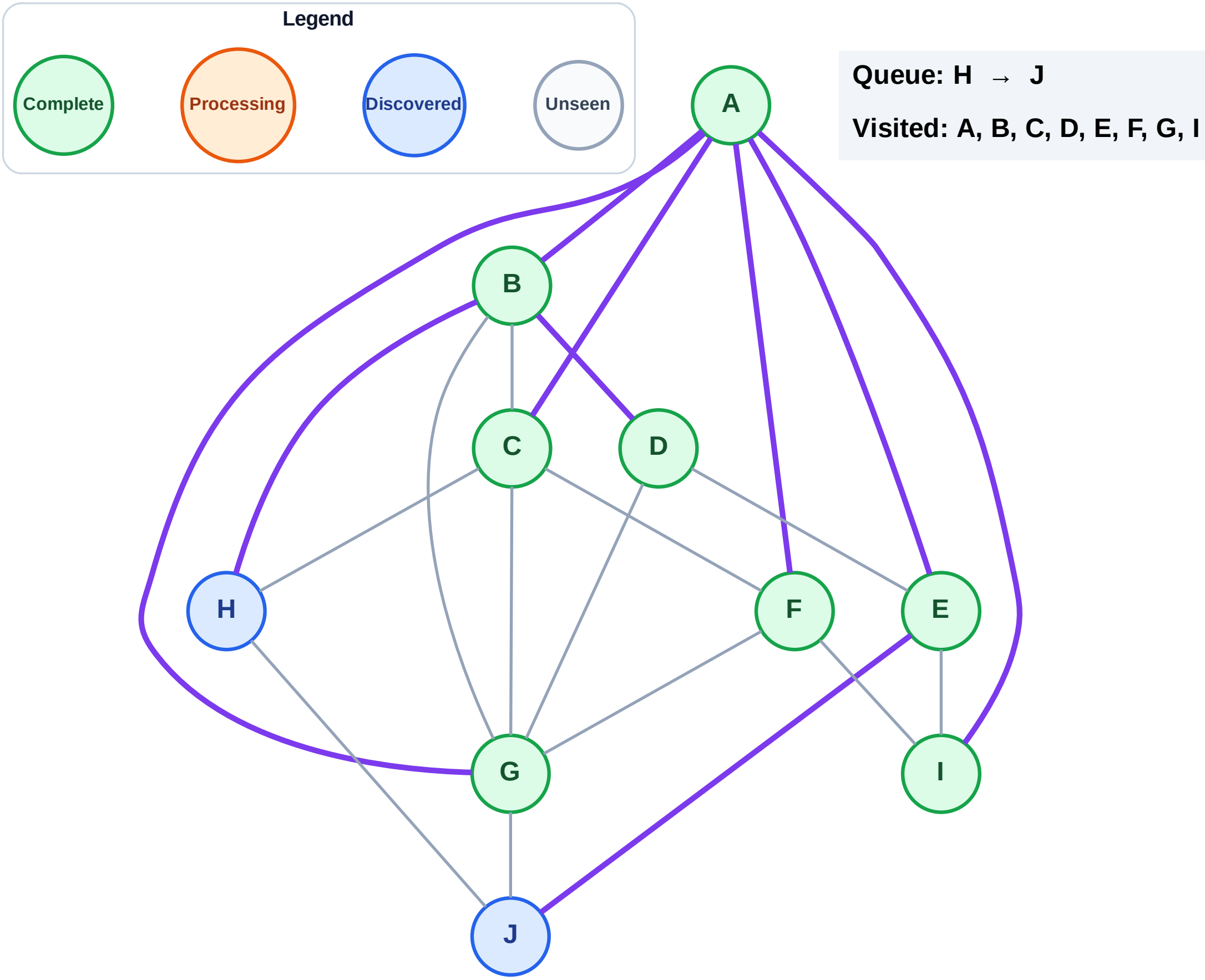


Queue: H → J

Visited: A, B, C, E, F, G, I

BFS Traversal

Step 17: No new neighbors from D

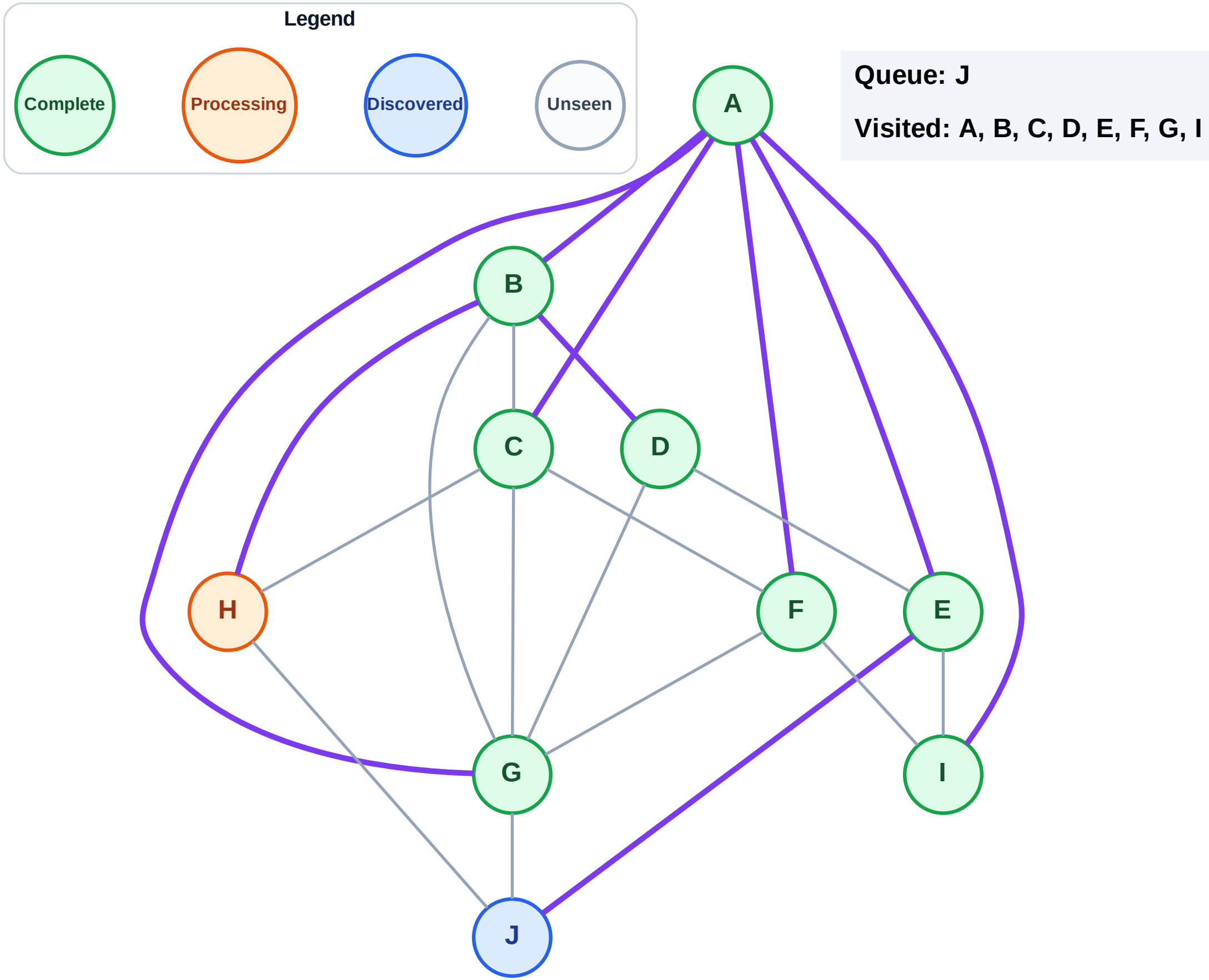


Queue: H → J

Visited: A, B, C, D, E, F, G, I

BFS Traversal

Step 18: Process node H

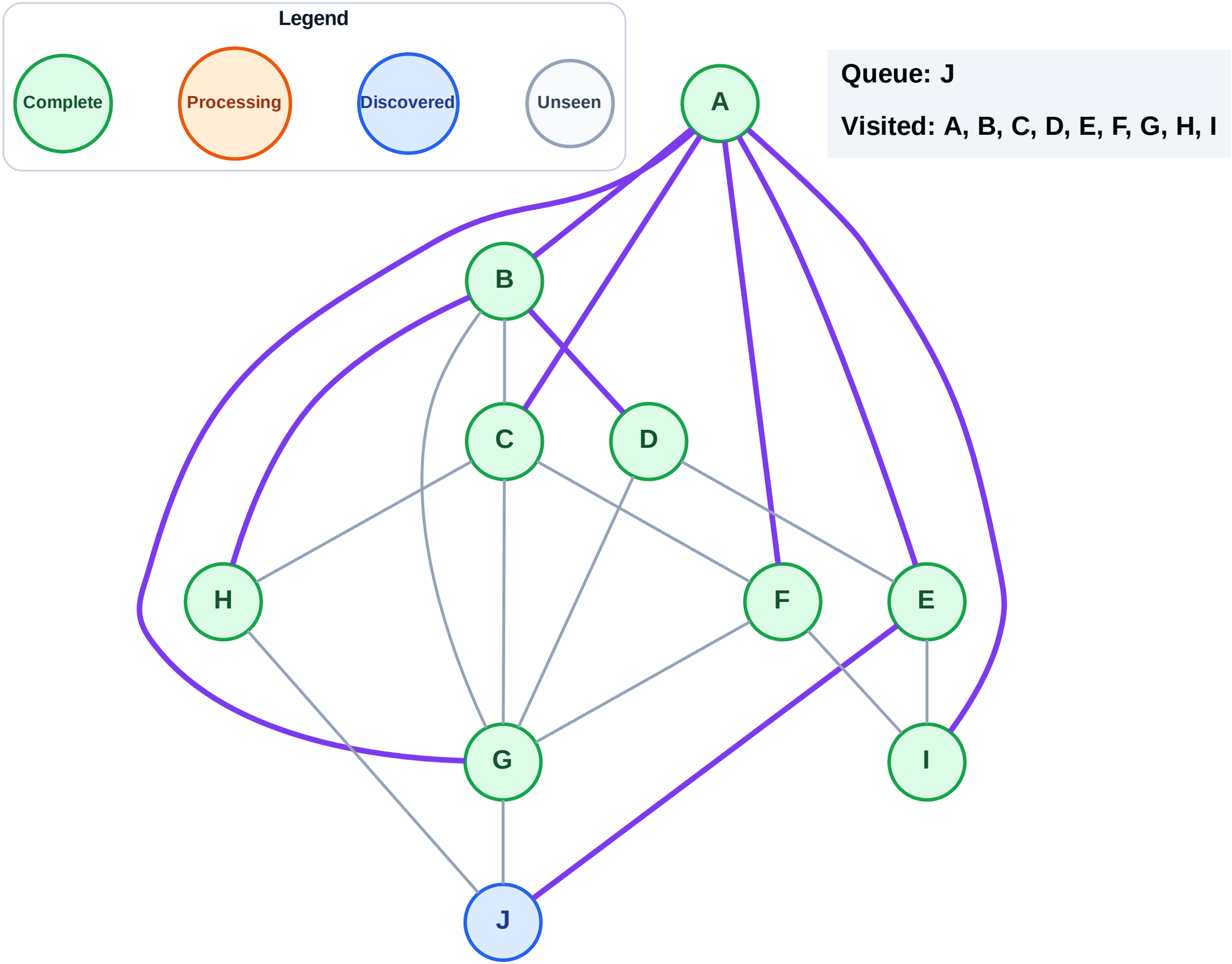


Queue: J

Visited: A, B, C, D, E, F, G, I

BFS Traversal

Step 19: No new neighbors from H

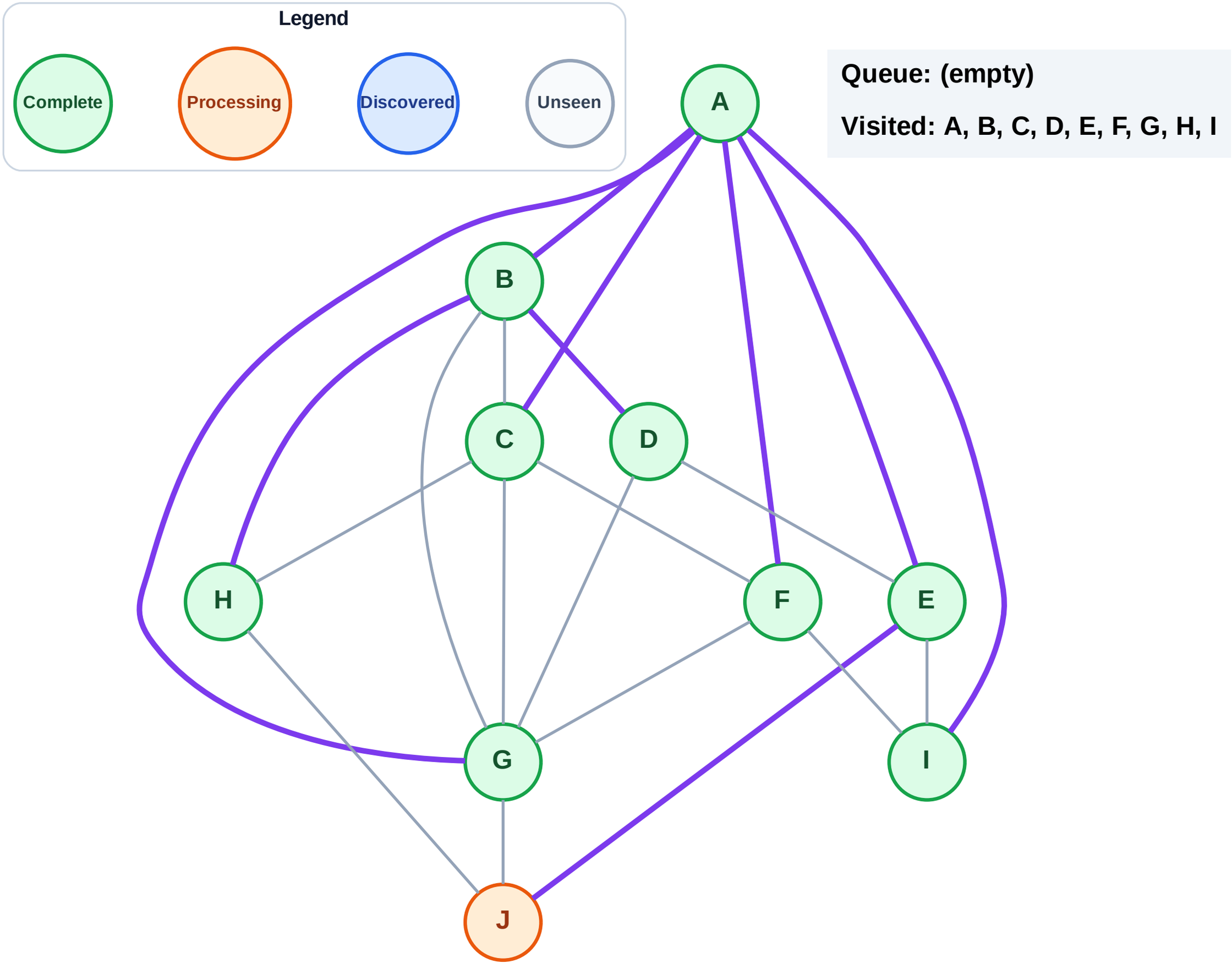


Queue: J

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 20: Process node J

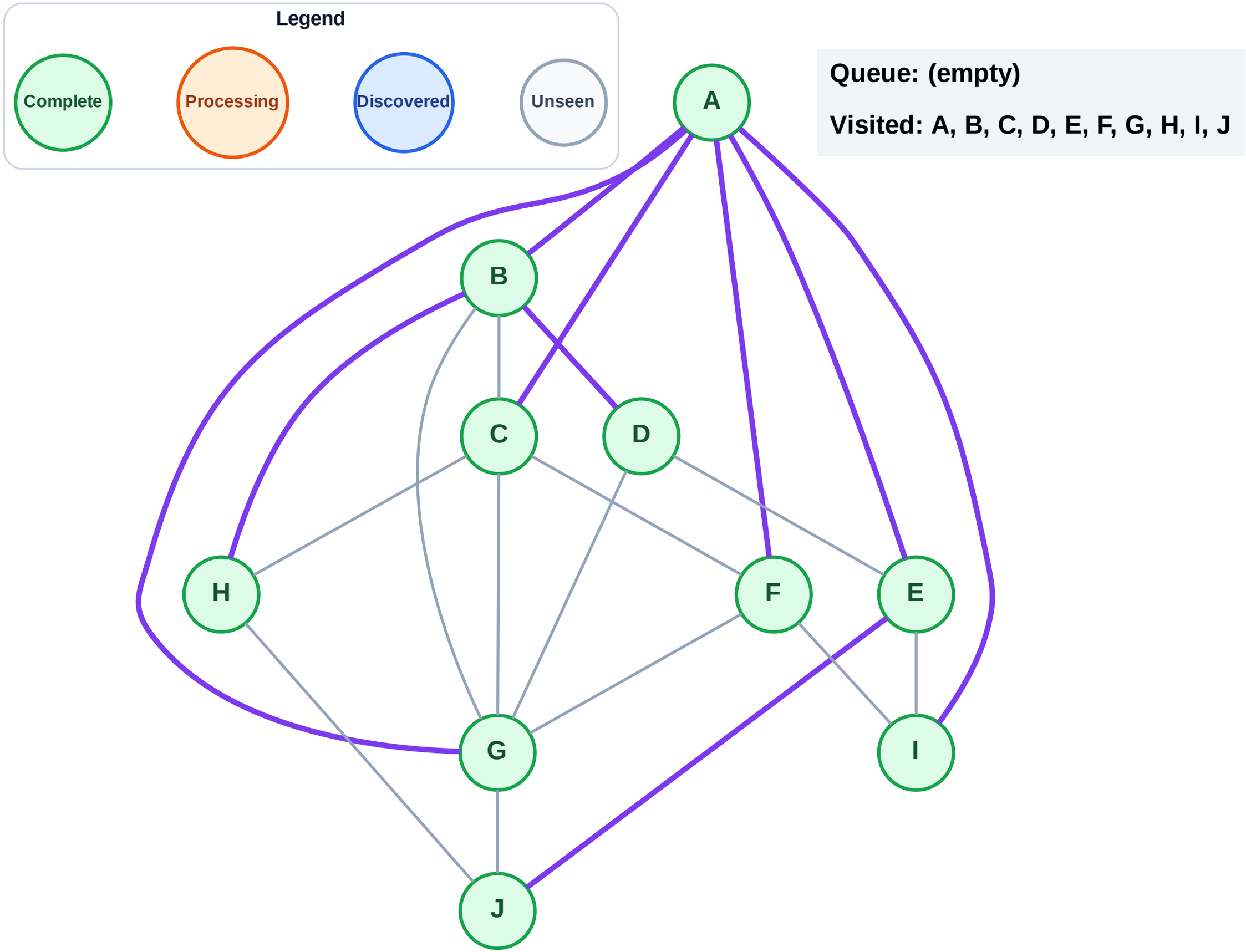


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 21: No new neighbors from J

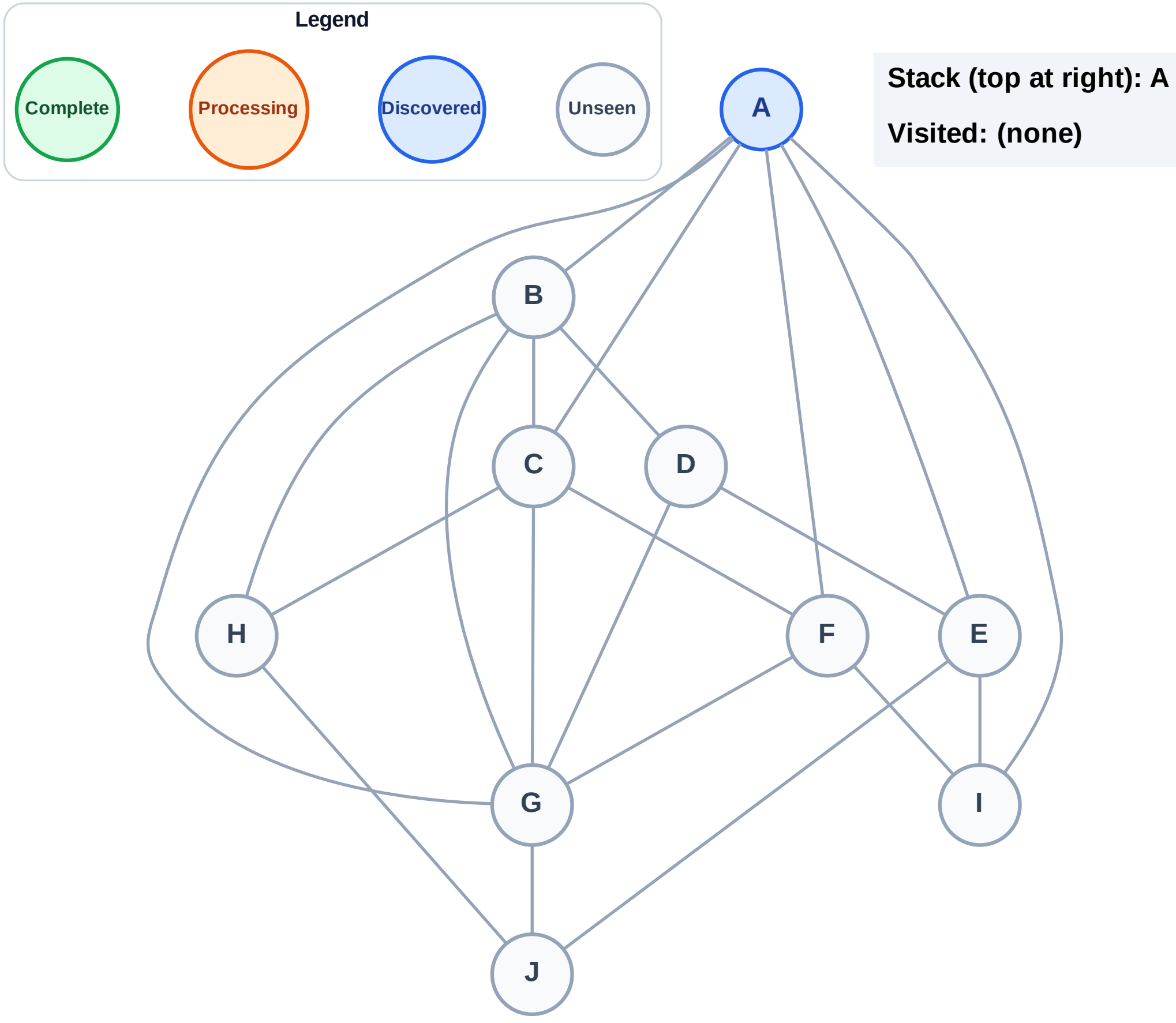


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

Step 1: Start at A

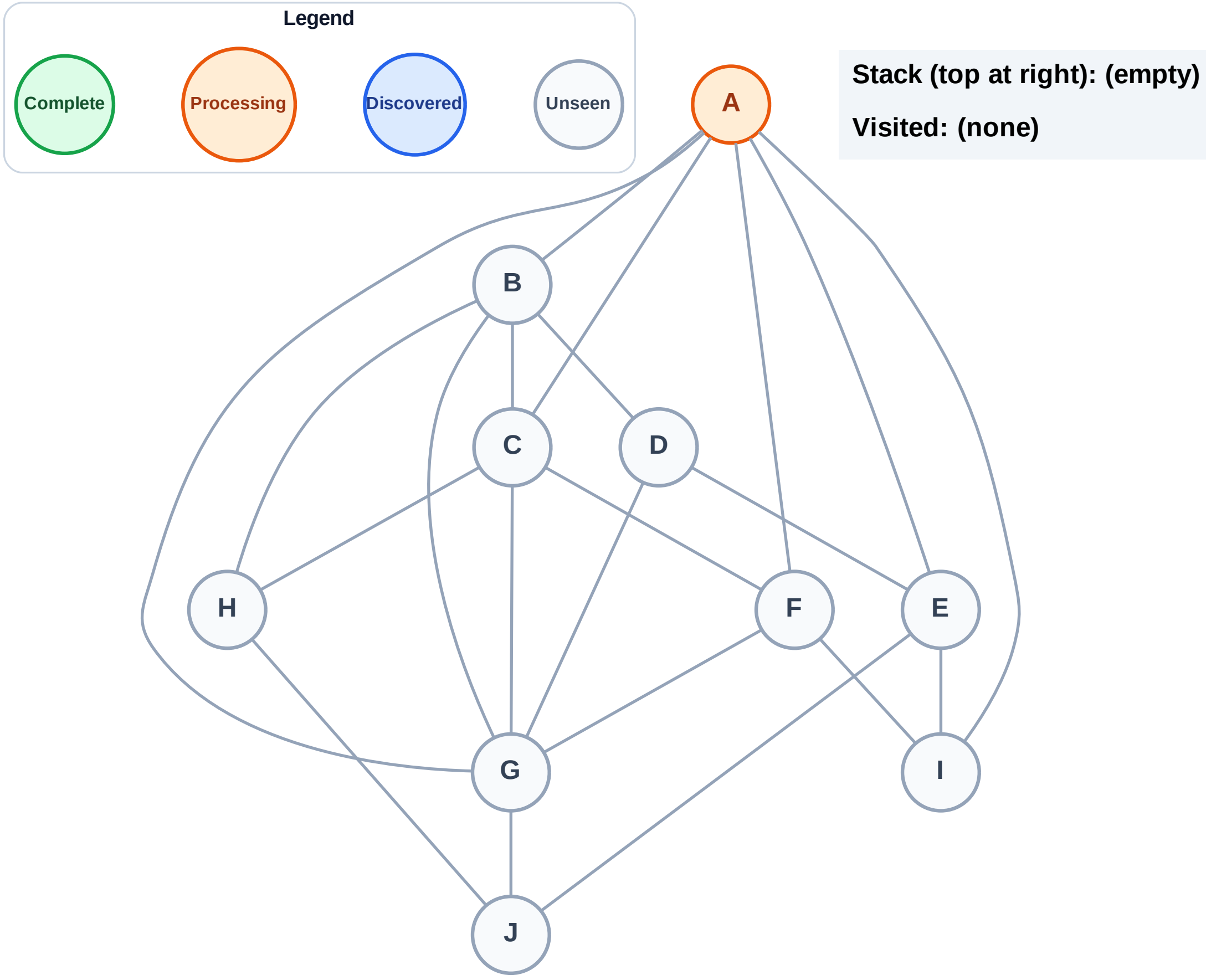


Stack (top at right): A

Visited: (none)

DFS Traversal

Step 2: Process node A

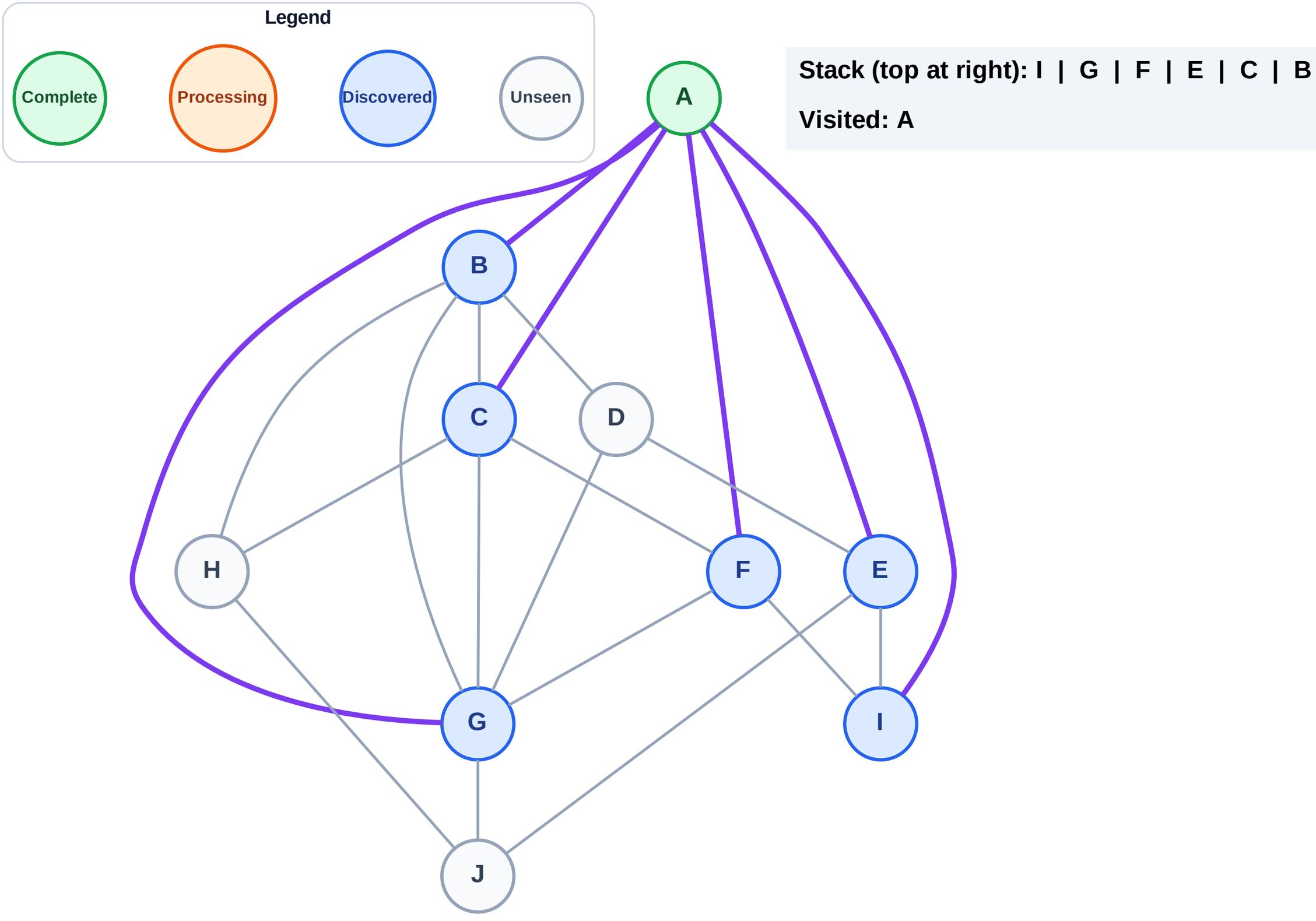


Stack (top at right): (empty)

Visited: (none)

DFS Traversal

Step 3: Discover I, G, F, E, C, B from A

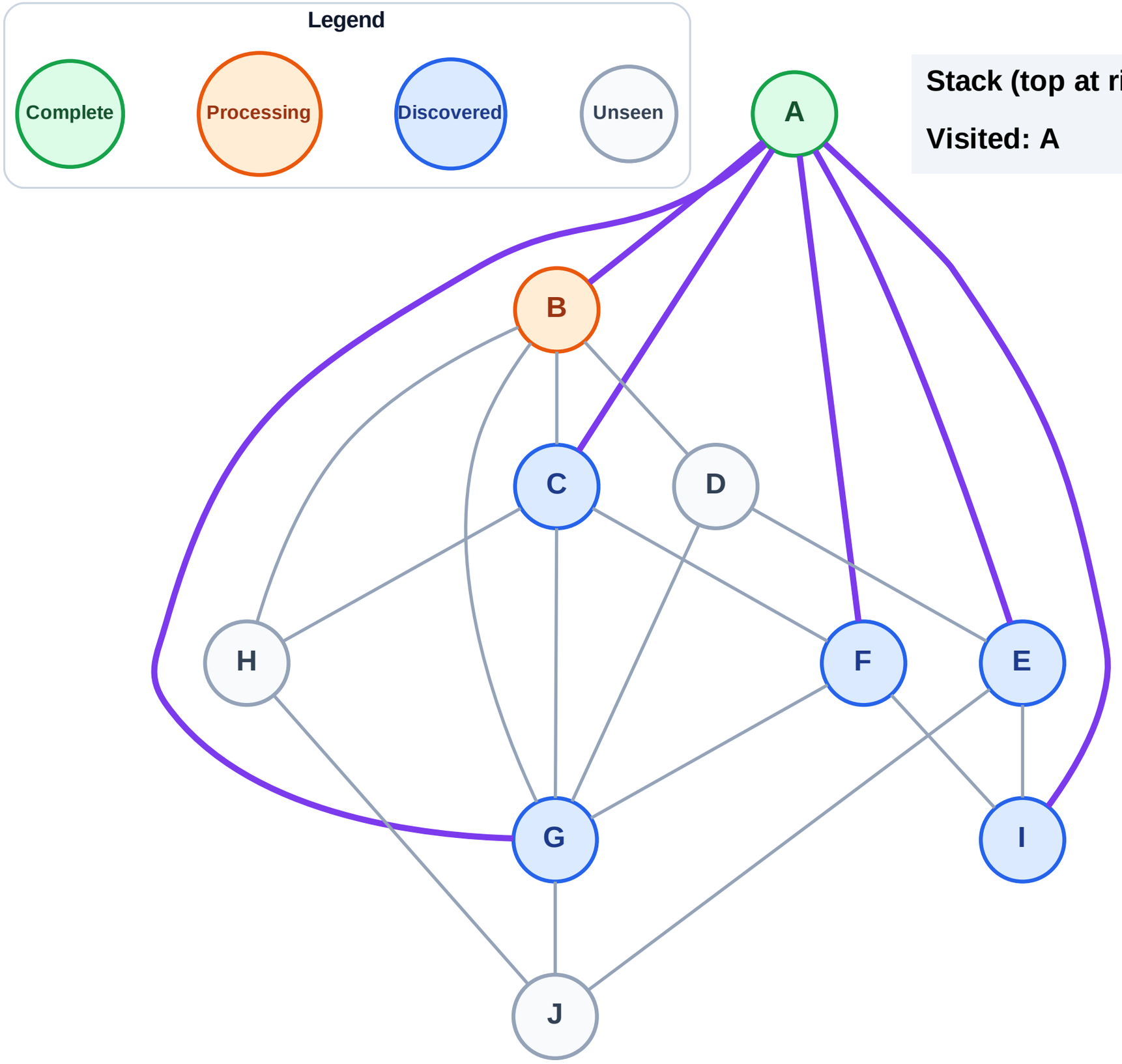


Stack (top at right): I | G | F | E | C | B

Visited: A

DFS Traversal

Step 4: Process node B

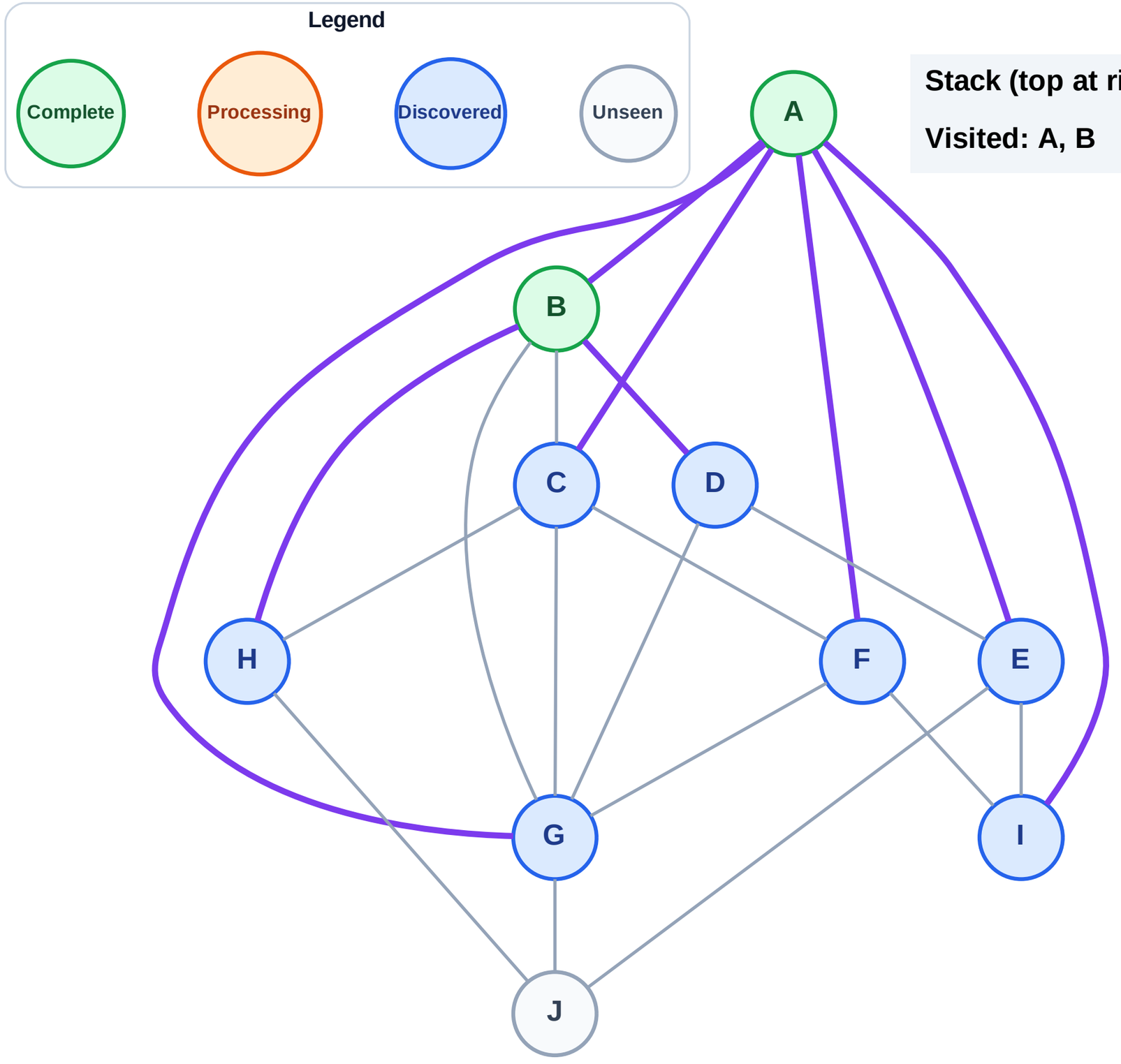


Stack (top at right): I | G | F | E | C

Visited: A

DFS Traversal

Step 5: Discover H, D from B

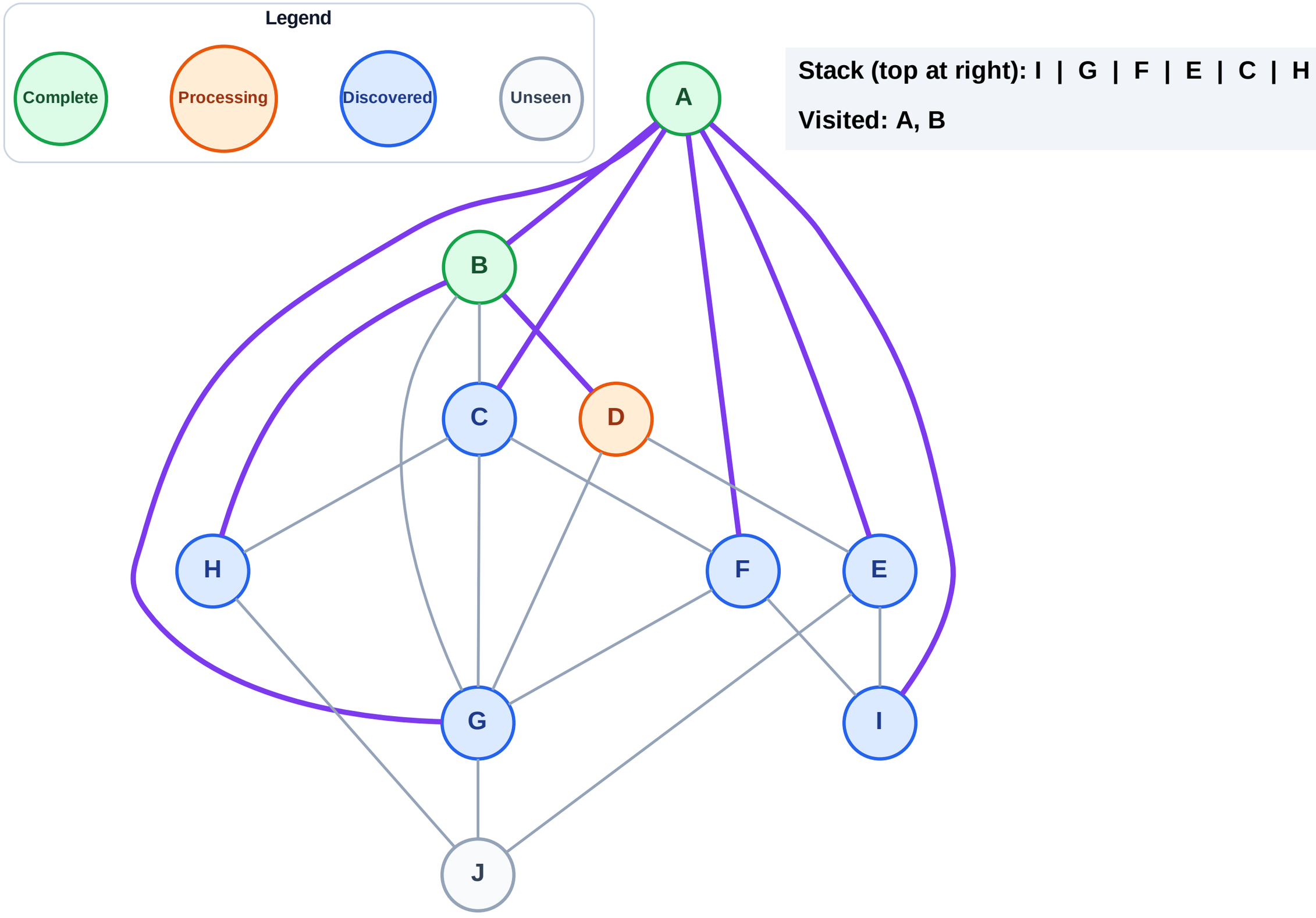


Stack (top at right): I | G | F | E | C | H | D

Visited: A, B

DFS Traversal

Step 6: Process node D

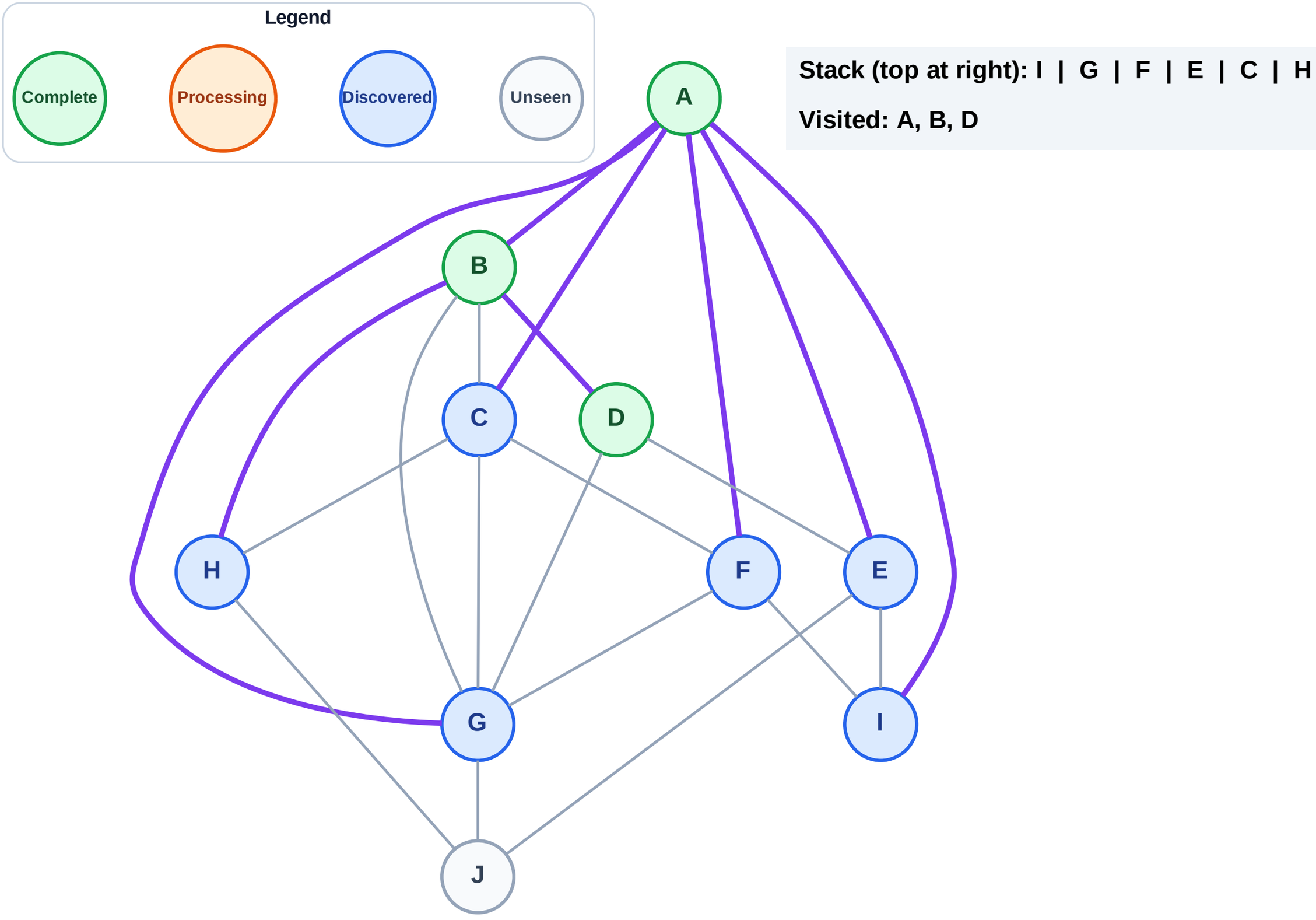


Stack (top at right): I | G | F | E | C | H

Visited: A, B

DFS Traversal

Step 7: No new neighbors from D

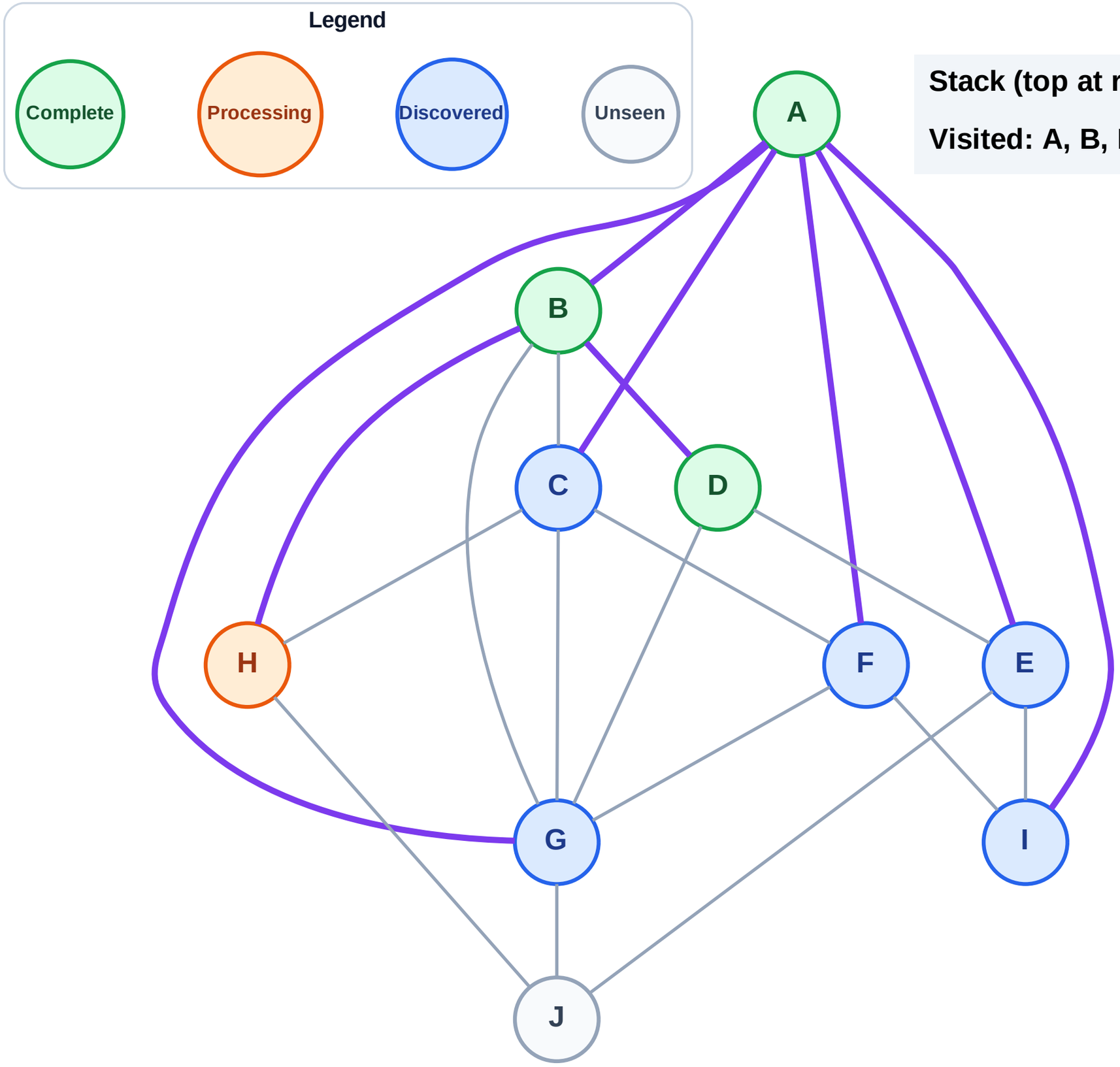


Stack (top at right): I | G | F | E | C | H

Visited: A, B, D

DFS Traversal

Step 8: Process node H

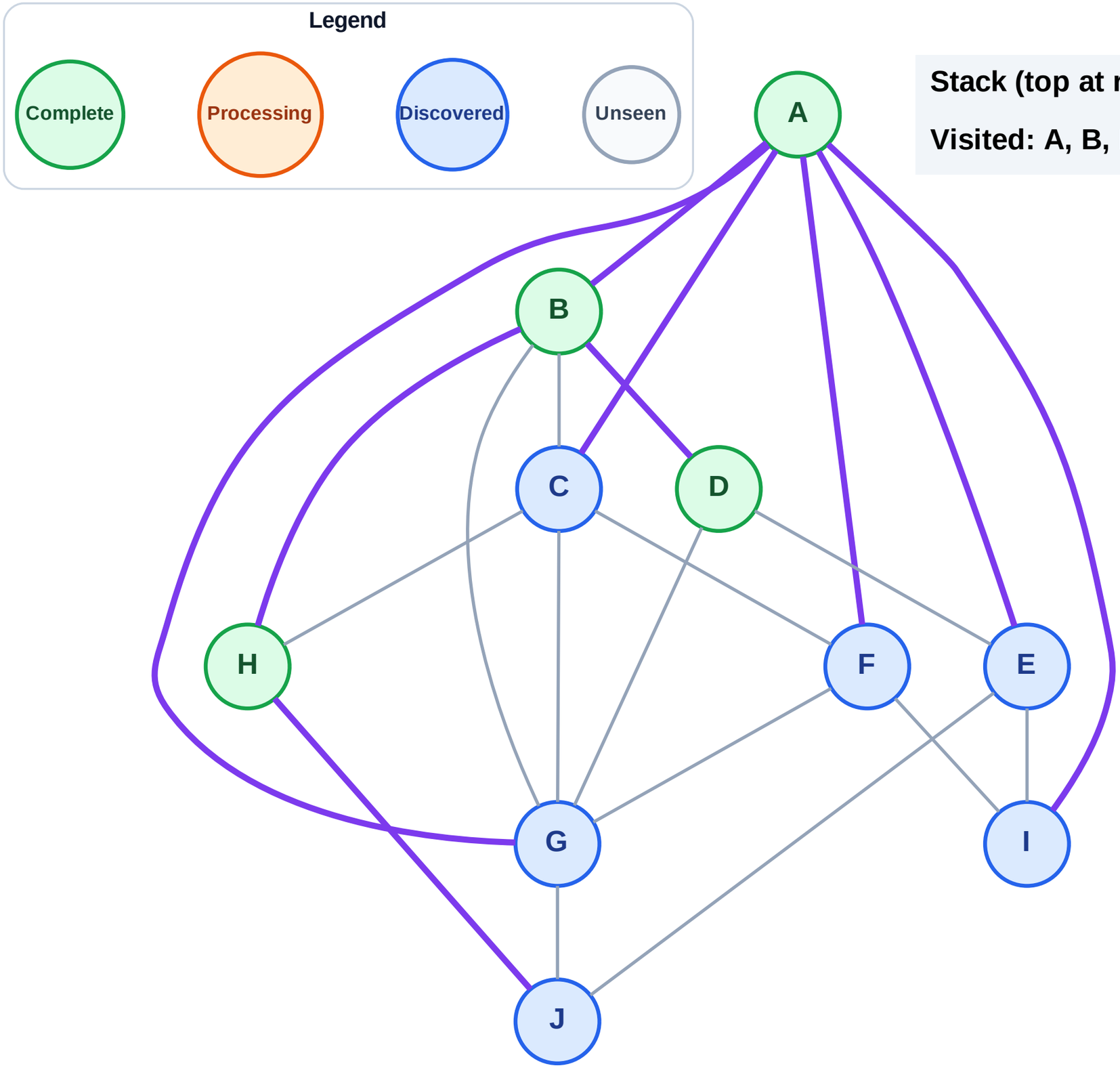


Stack (top at right): I | G | F | E | C

Visited: A, B, D

DFS Traversal

Step 9: Discover J from H

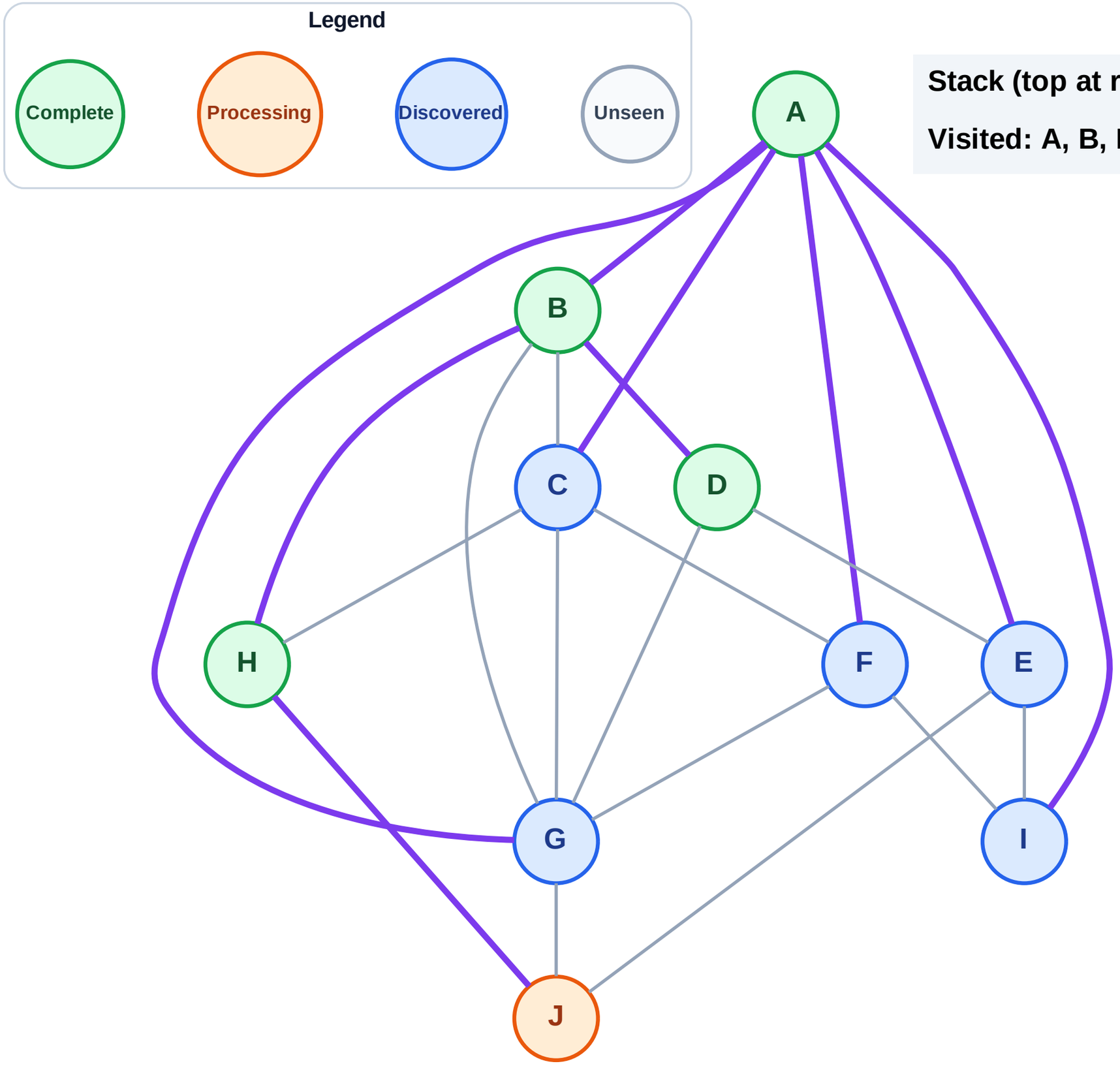


Stack (top at right): I | G | F | E | C | J

Visited: A, B, D, H

DFS Traversal

Step 10: Process node J

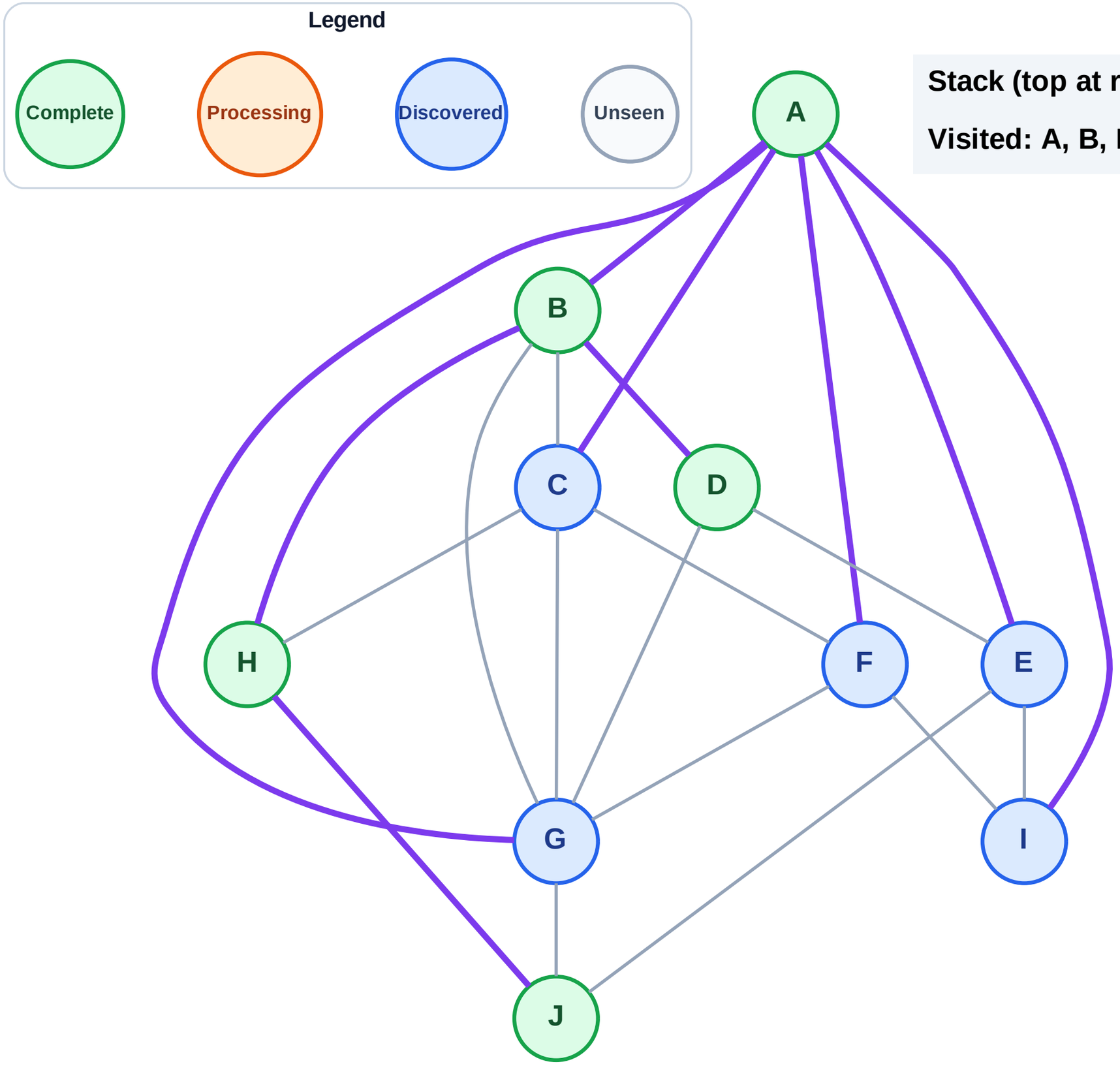


Stack (top at right): I | G | F | E | C

Visited: A, B, D, H

DFS Traversal

Step 11: No new neighbors from J

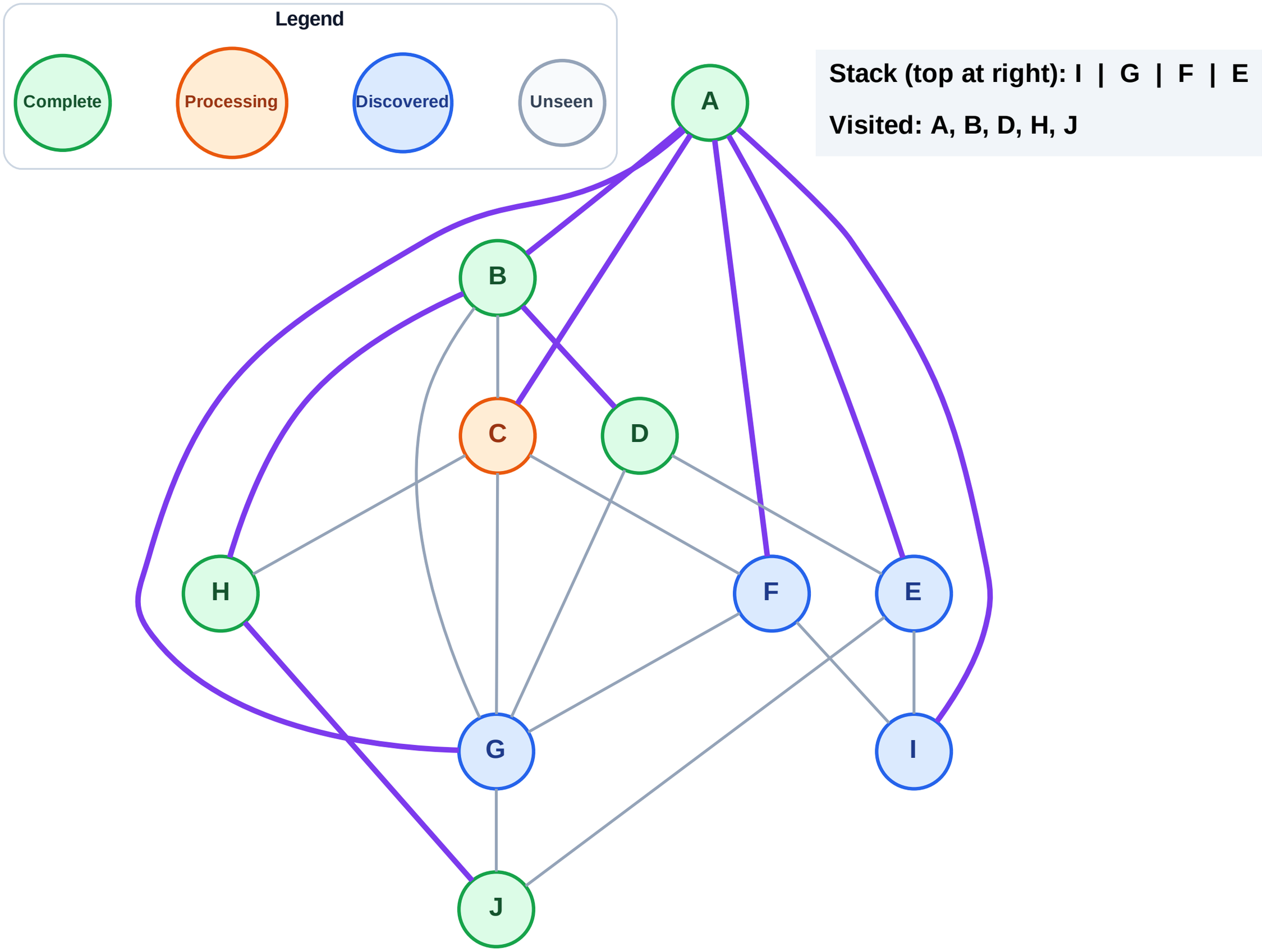


Stack (top at right): I | G | F | E | C

Visited: A, B, D, H, J

DFS Traversal

Step 12: Process node C

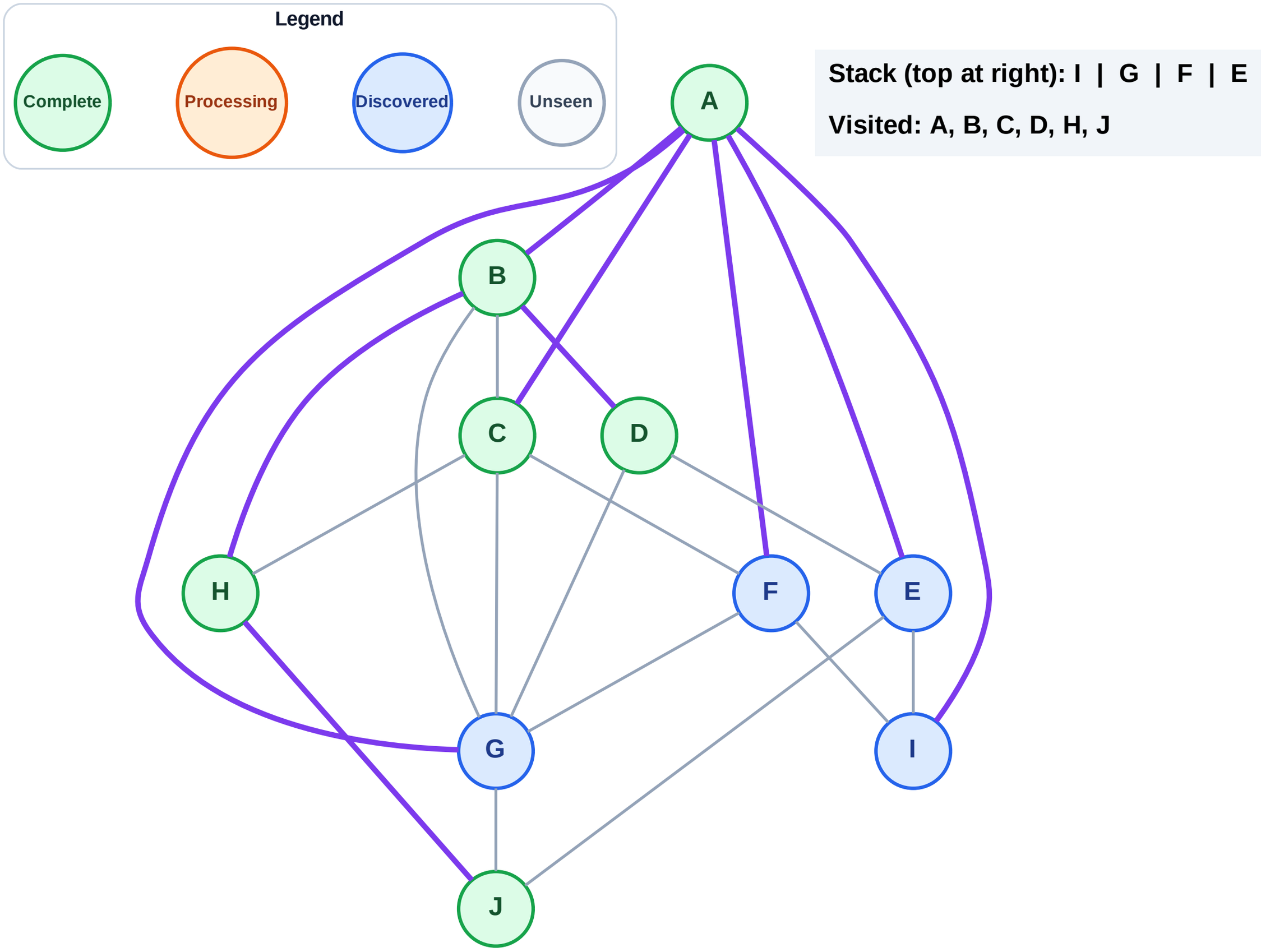


Stack (top at right): I | G | F | E

Visited: A, B, D, H, J

DFS Traversal

Step 13: No new neighbors from C

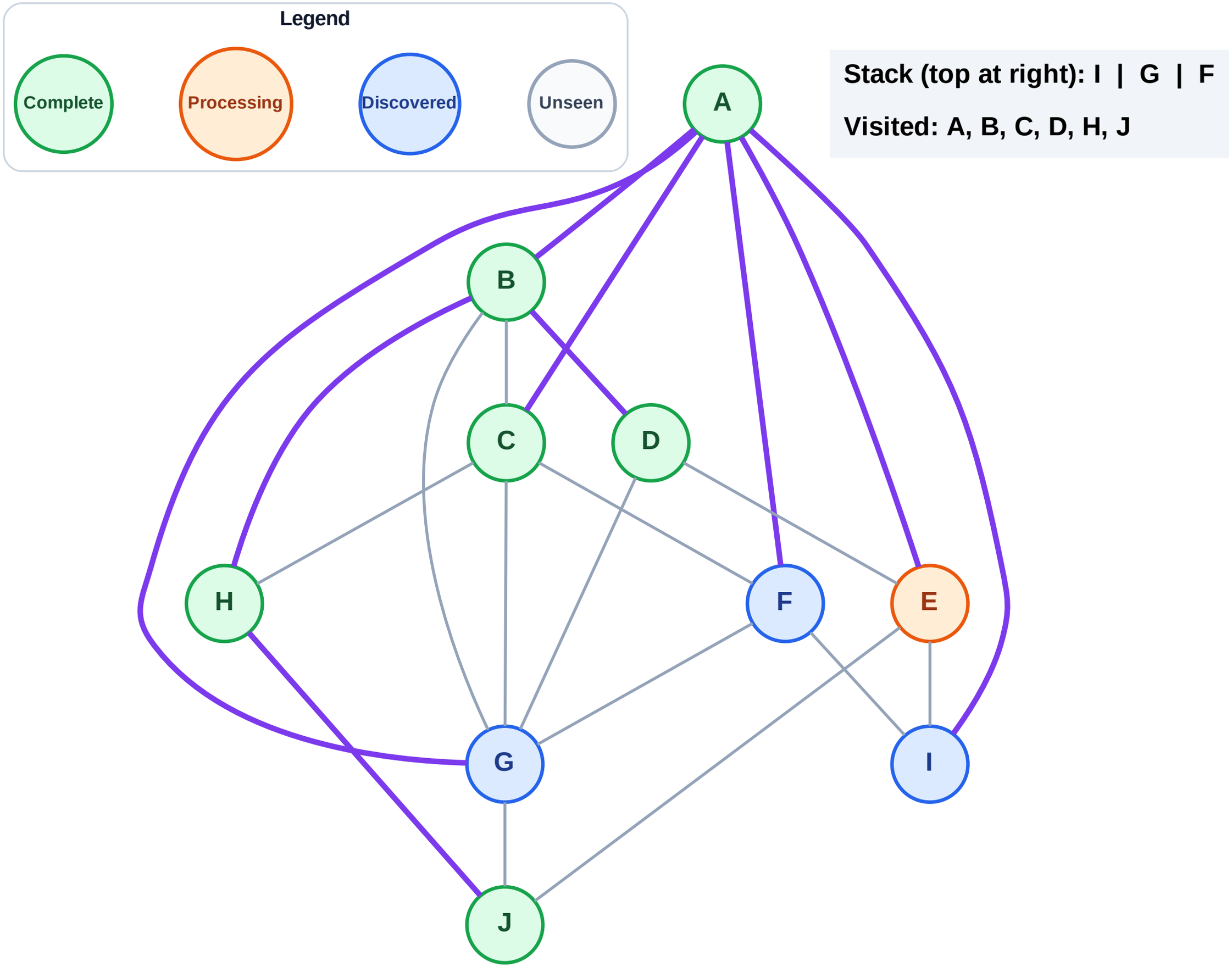


Stack (top at right): I | G | F | E

Visited: A, B, C, D, H, J

DFS Traversal

Step 14: Process node E

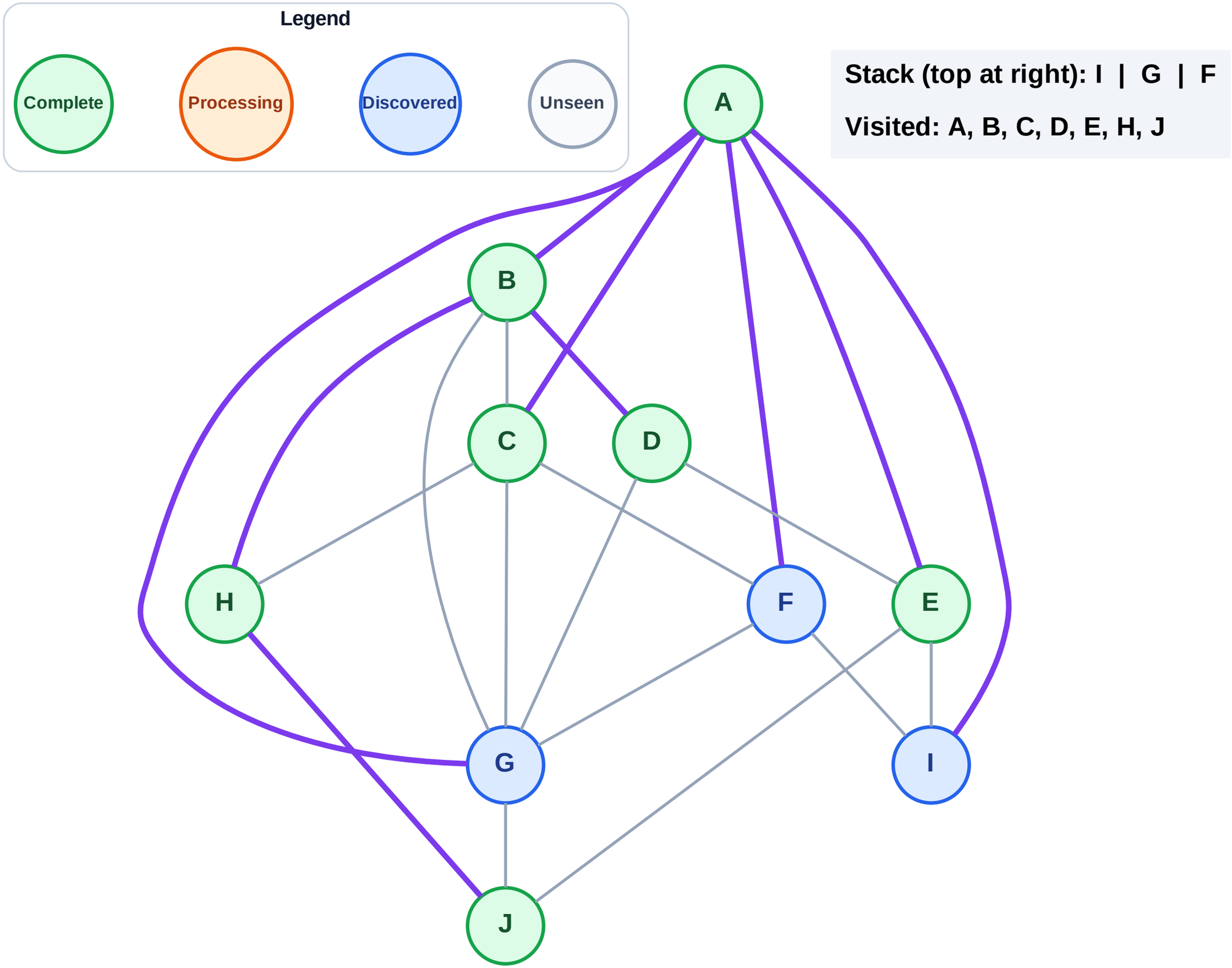


Stack (top at right): I | G | F

Visited: A, B, C, D, H, J

DFS Traversal

Step 15: No new neighbors from E

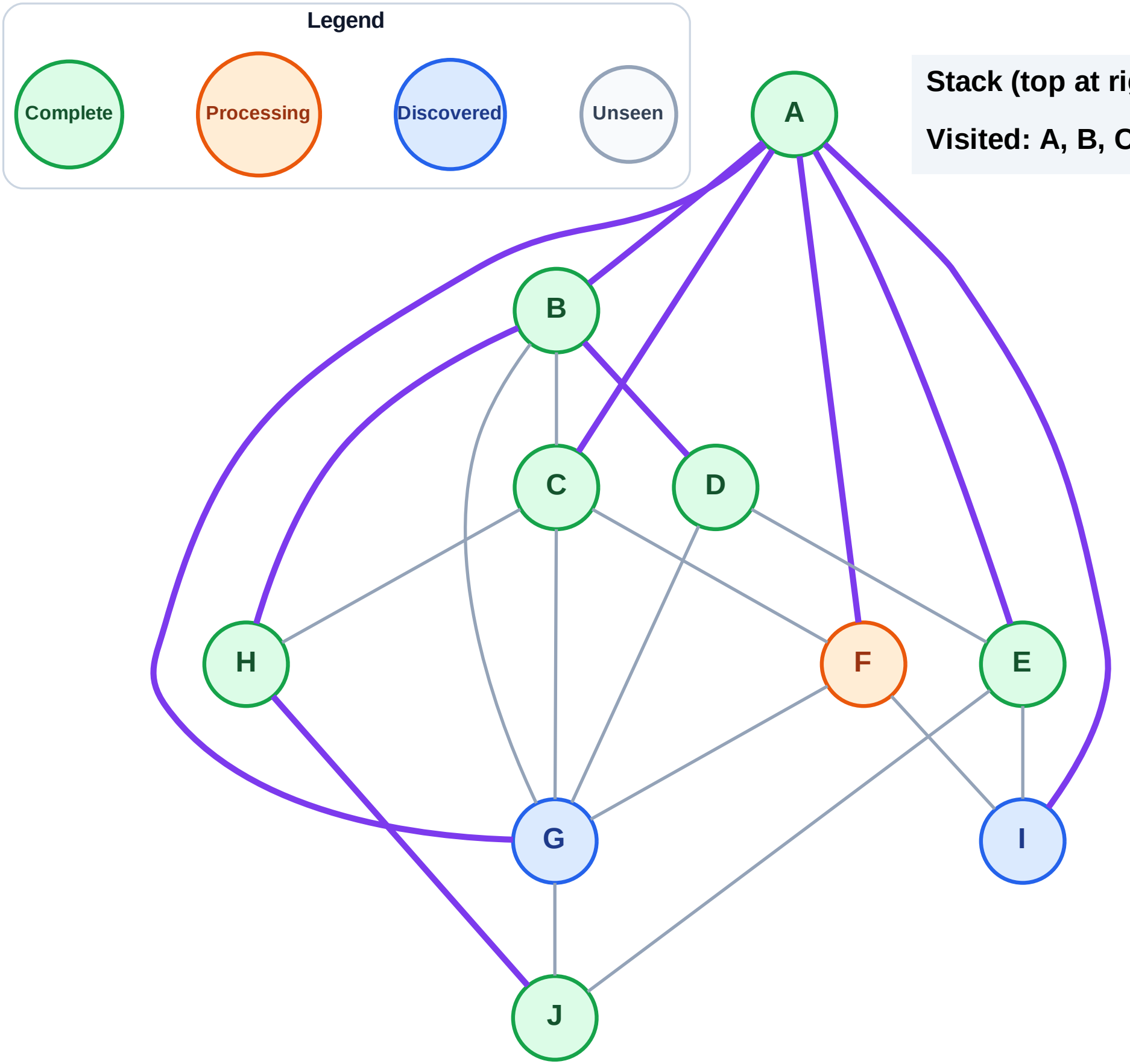


Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J

DFS Traversal

Step 16: Process node F

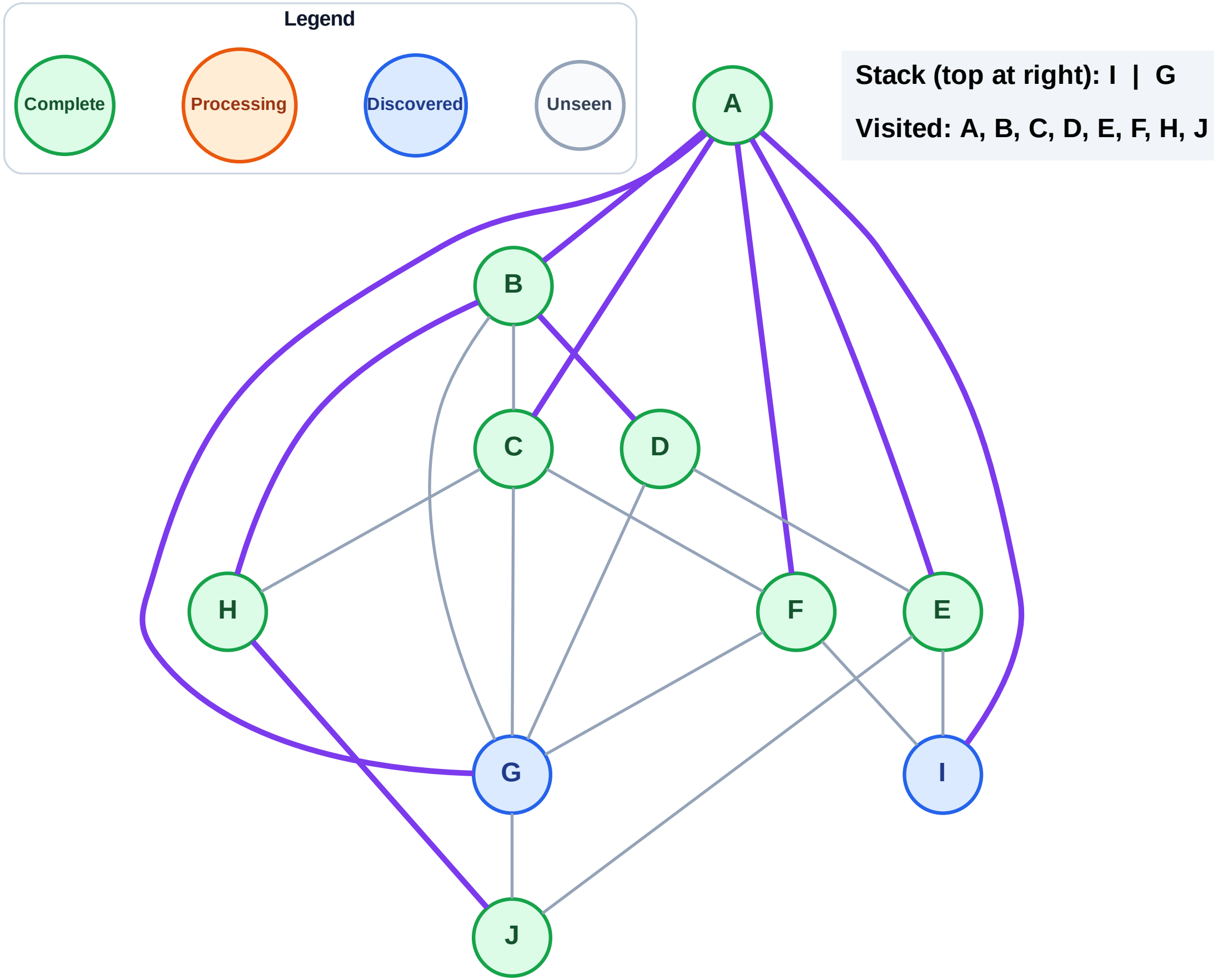


Stack (top at right): I | G

Visited: A, B, C, D, E, H, J

DFS Traversal

Step 17: No new neighbors from F

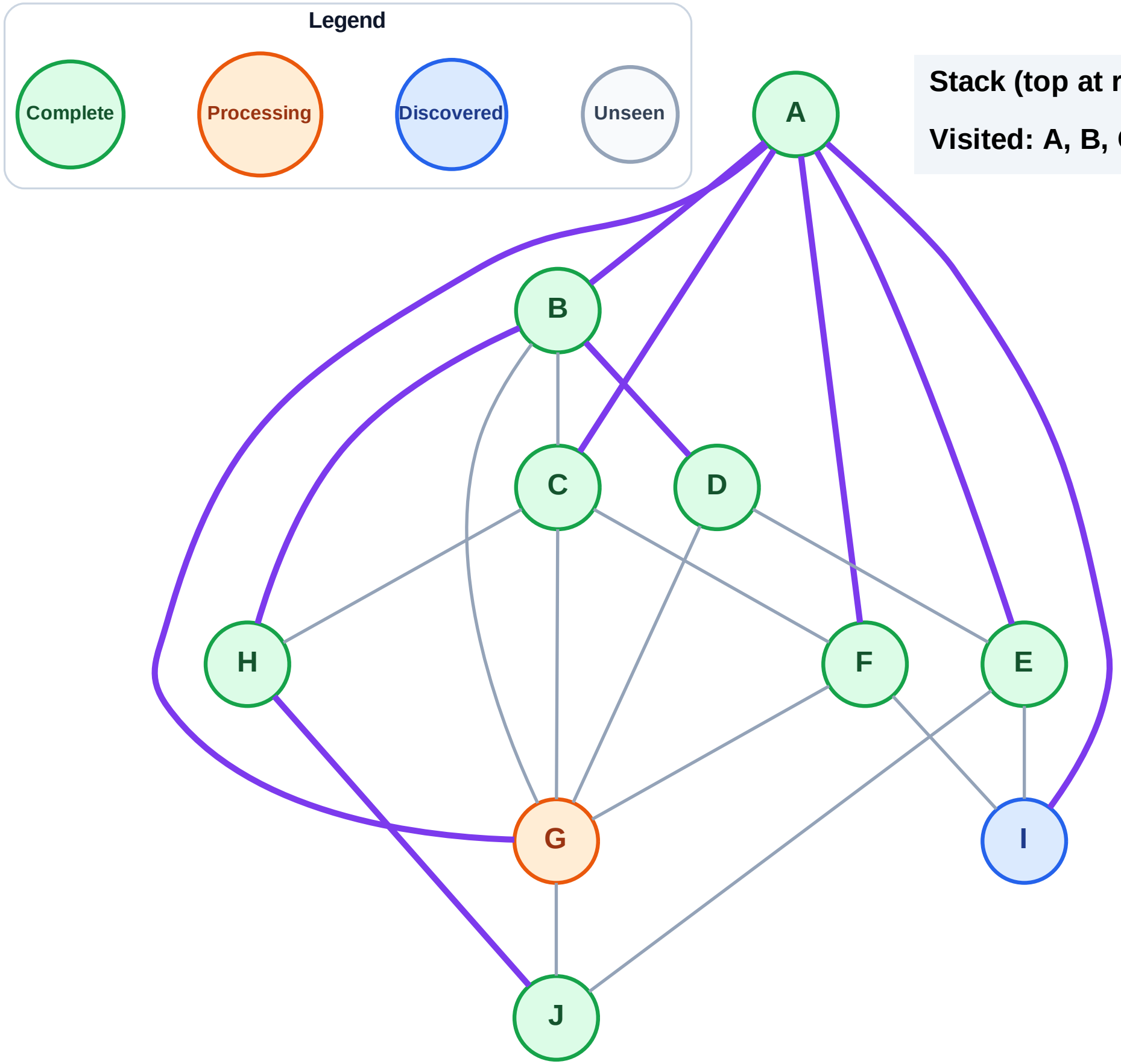


Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J

DFS Traversal

Step 18: Process node G

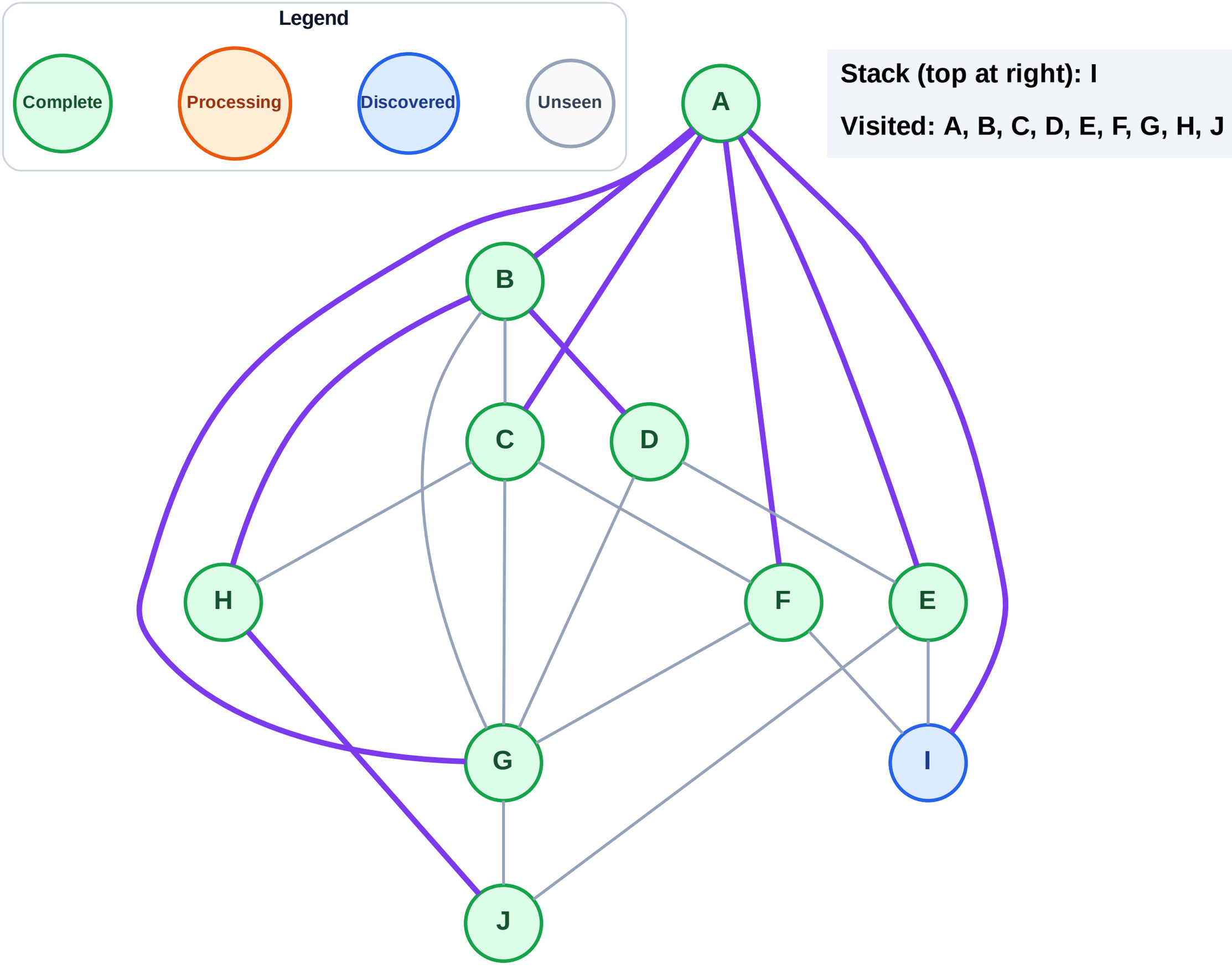


Stack (top at right): I

Visited: A, B, C, D, E, F, H, J

DFS Traversal

Step 19: No new neighbors from G

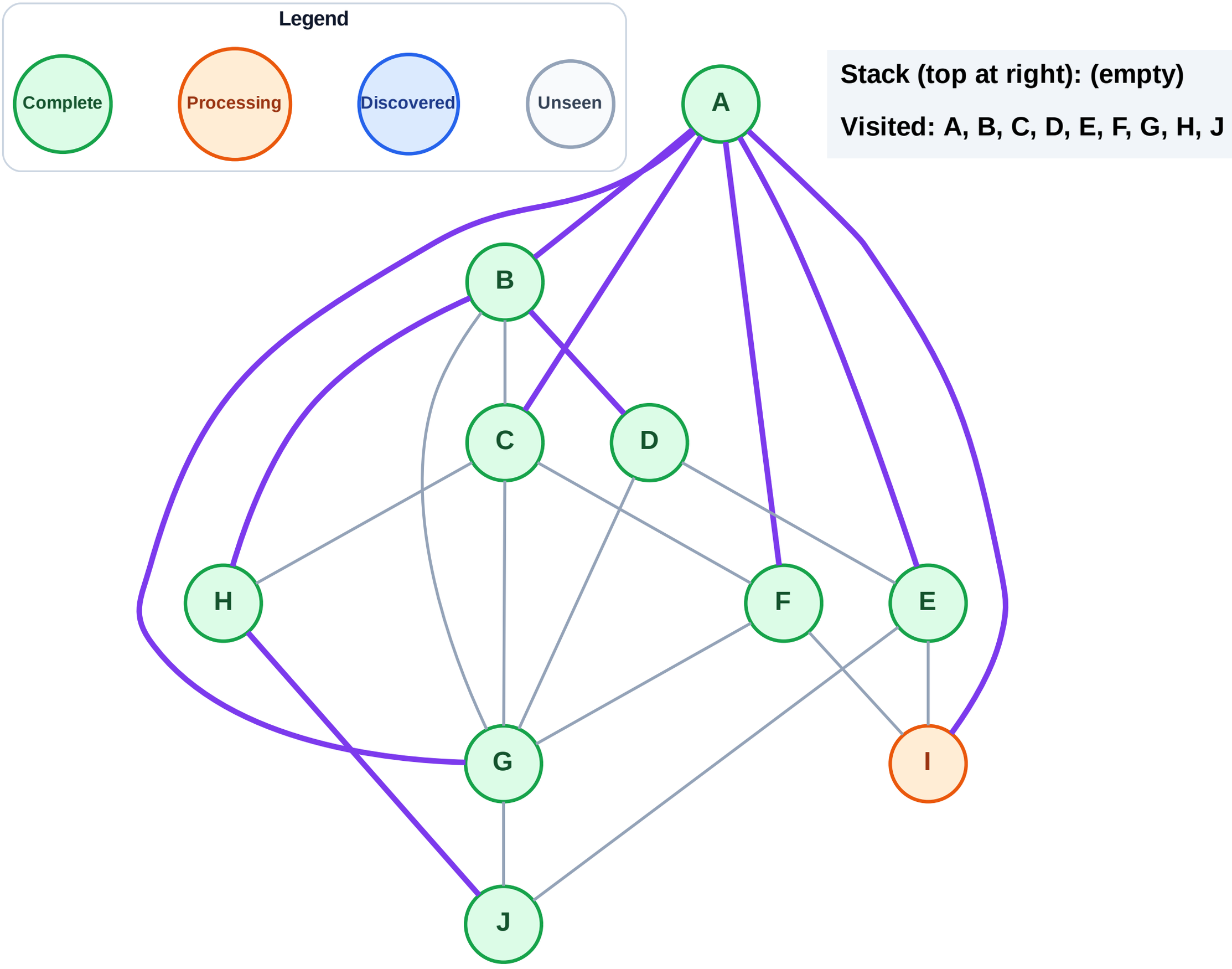


Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J

DFS Traversal

Step 20: Process node I

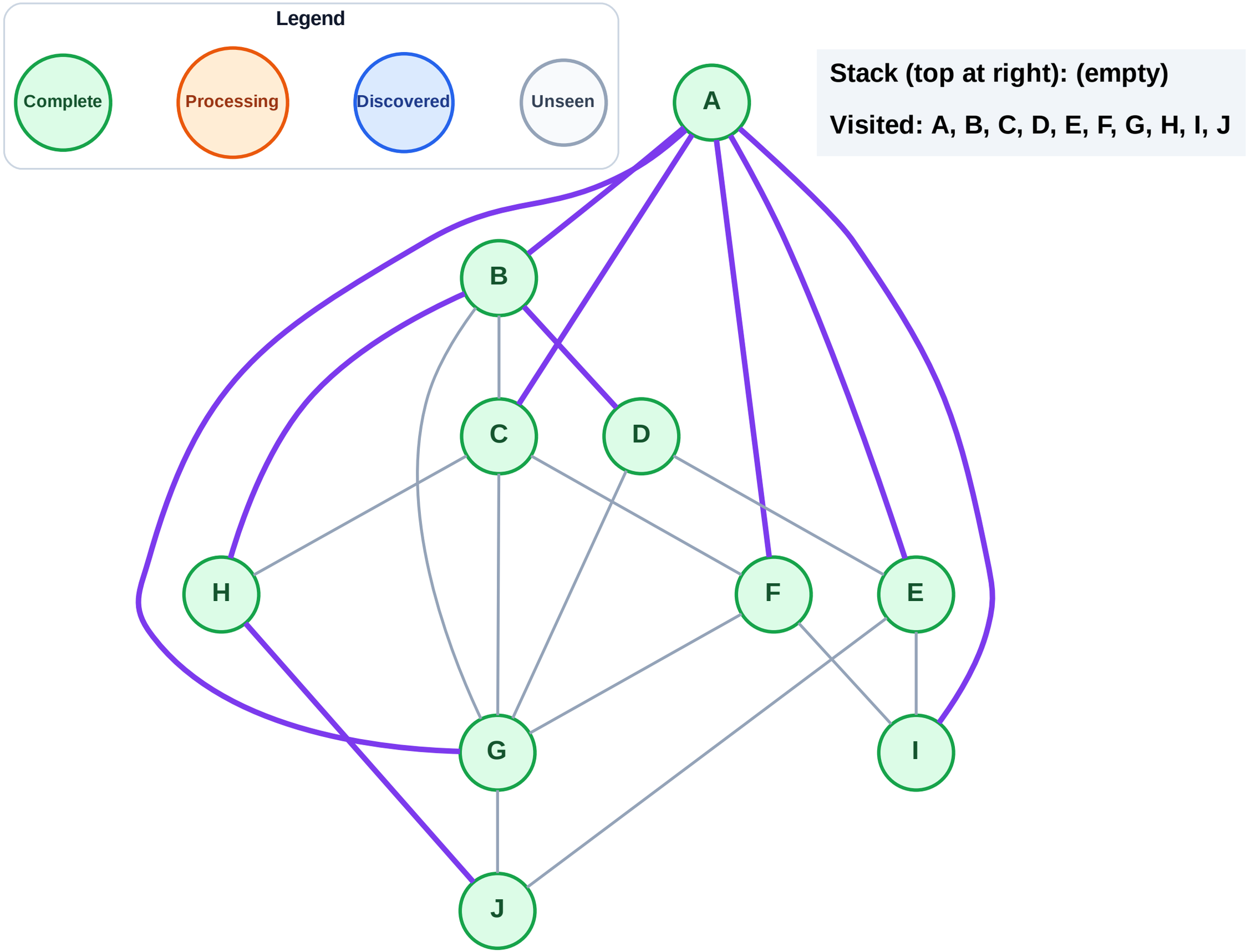


Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J

DFS Traversal

Step 21: No new neighbors from I



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J